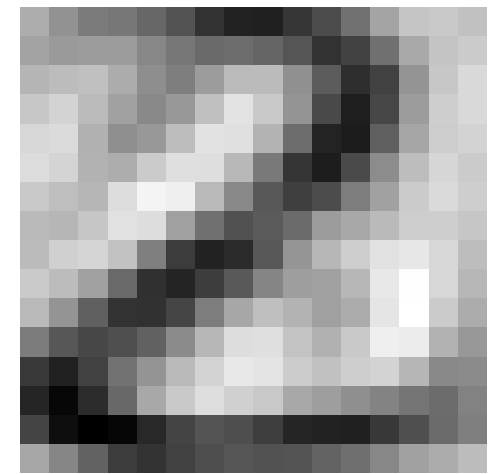
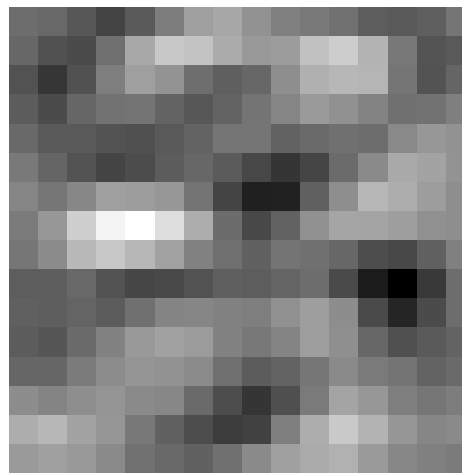
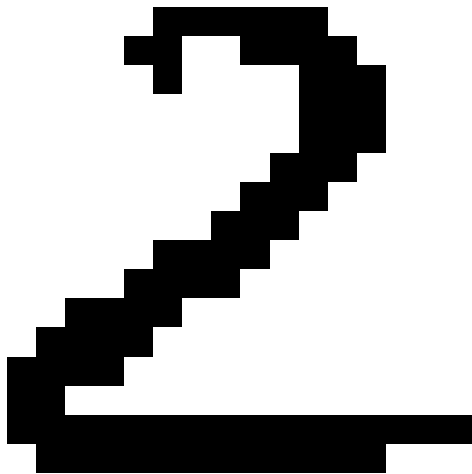


# Advanced Machine Learning

## *Principal Component Analysis (PCA)*

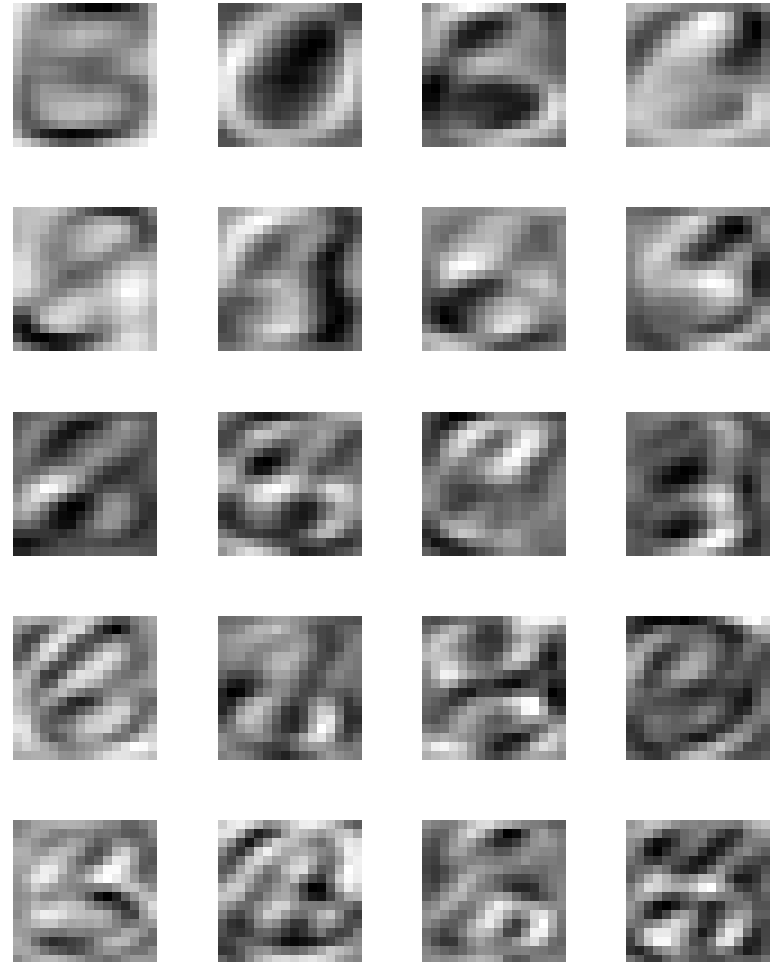
1.6 -1.1 -1.6 2.1 -0.52 2.8 0.72 0.7 -0.68 -0.41 -1.4 -1.5 -0.54 -0.62 1.3 -1.4 -0.27 0.74 0.77 -1



*Covariance matrices, dimensionality reduction, PCA, Duality*

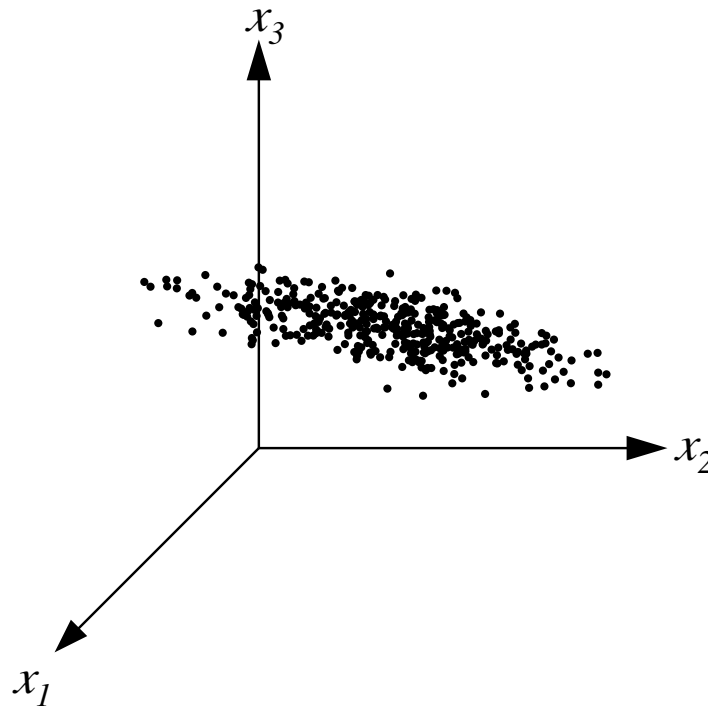
# Outline

1. **Covariance Matrices**
2. Principal Component Analysis
3. Duality



# Spread of Data

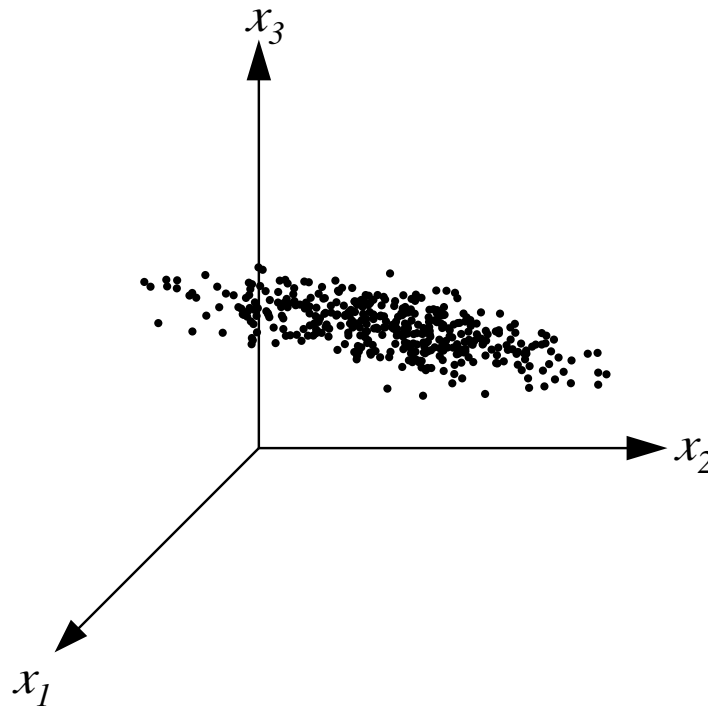
- Often data varies significantly in only some directions



- Reduce dimensions by projecting onto low dimensional subspace with maximum variation

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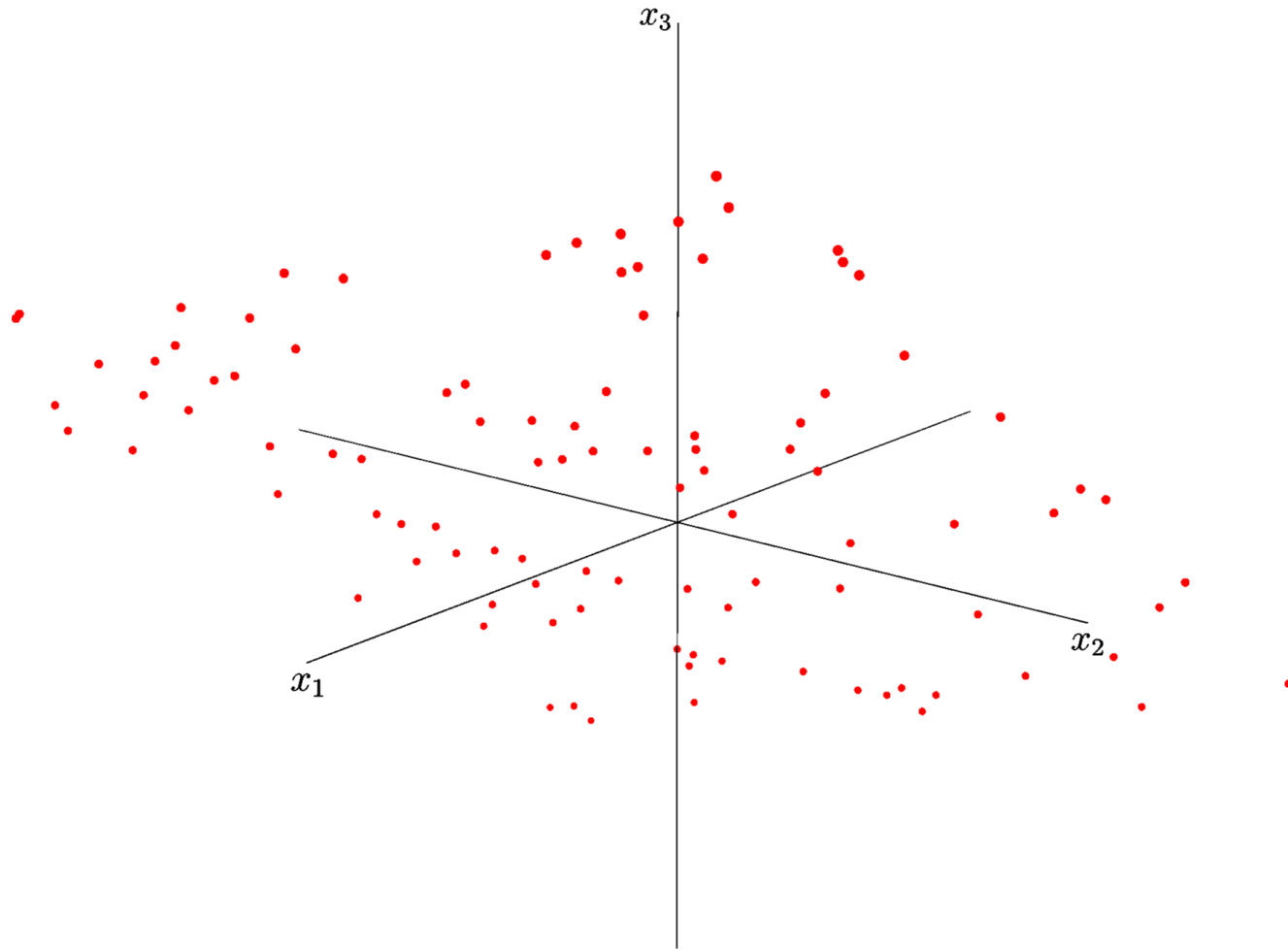
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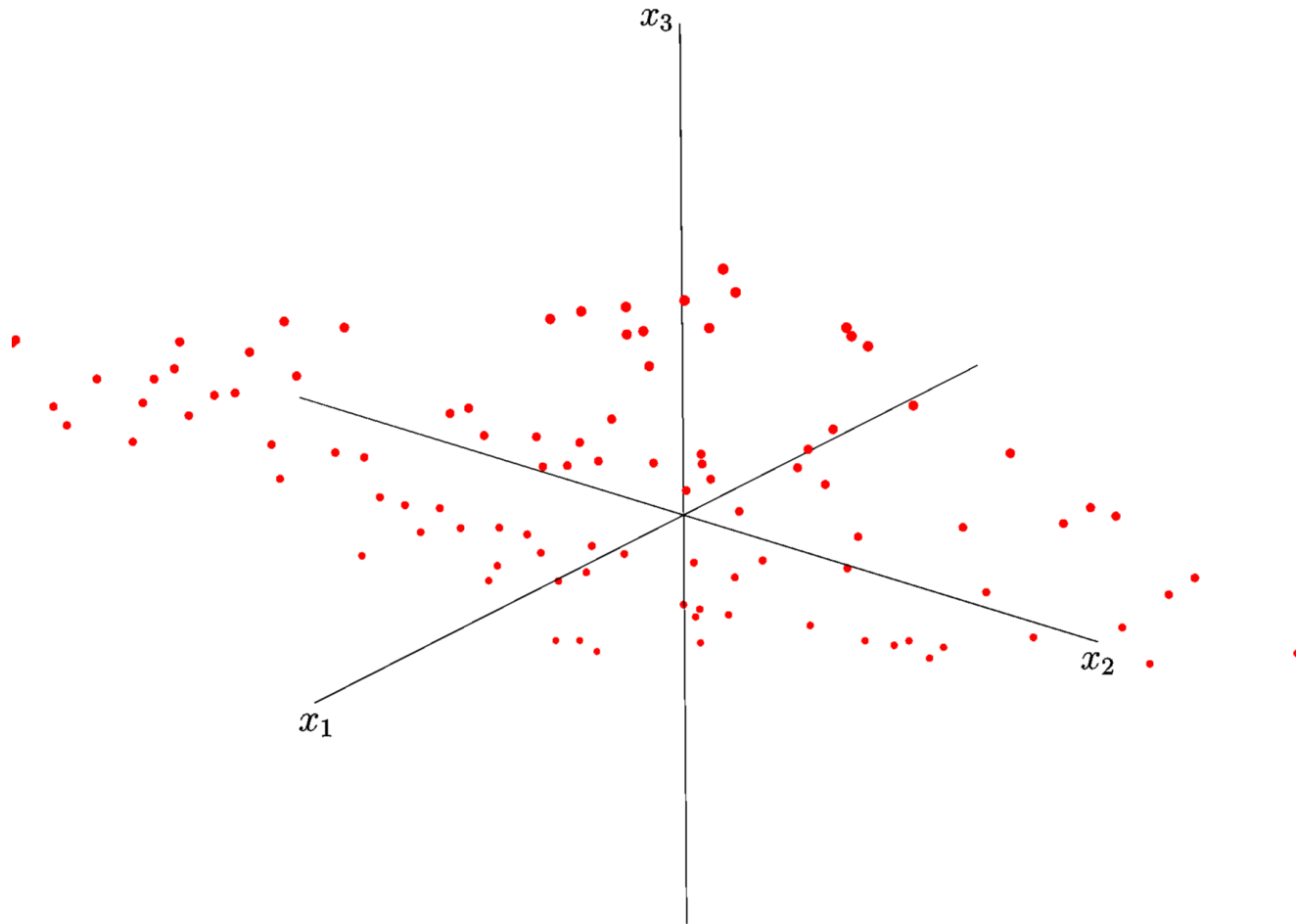
# Looking is not Enough

Can't spot low dimensional data by looking at numbers



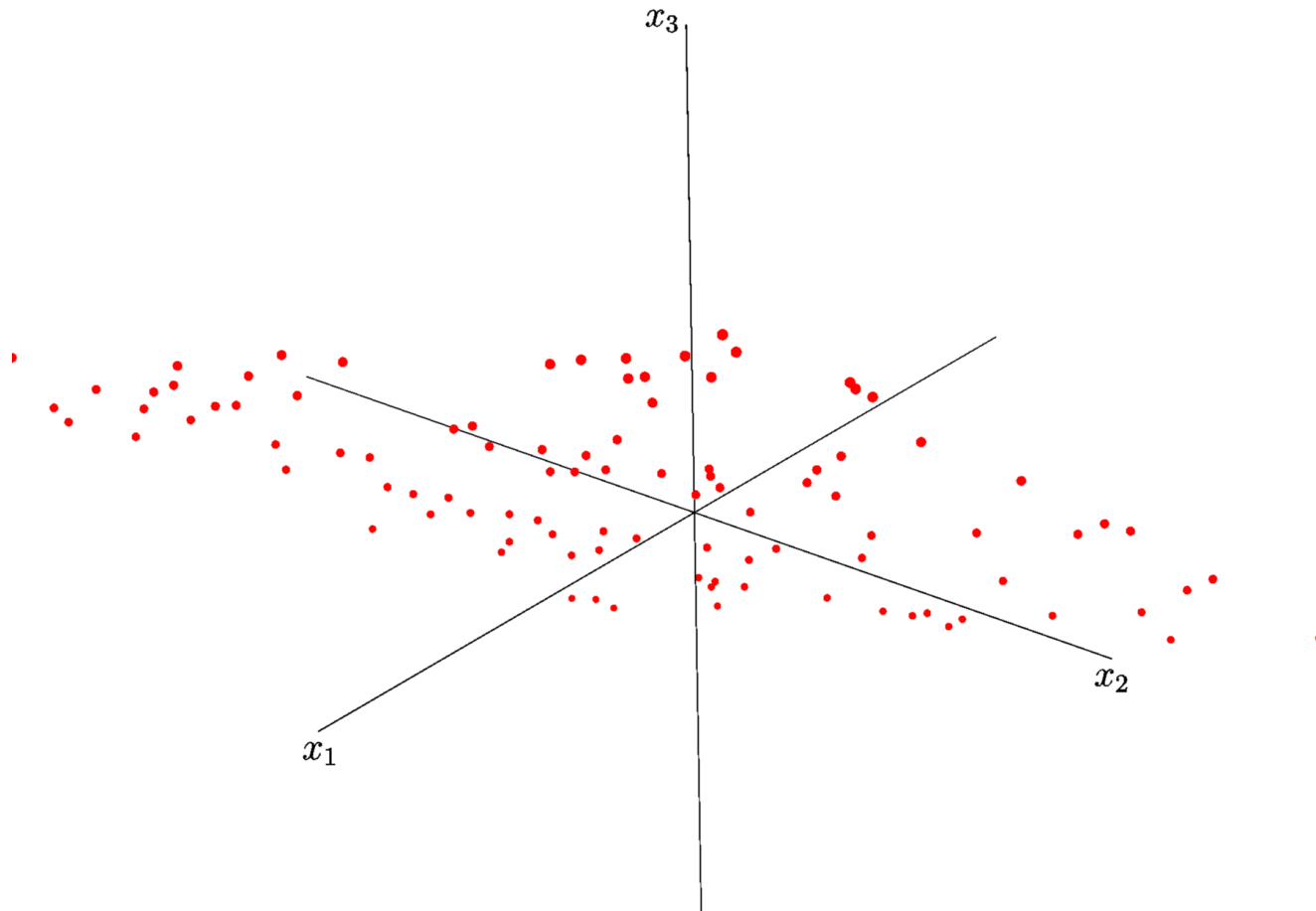
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Can't spot low dimensional data by looking at numbers



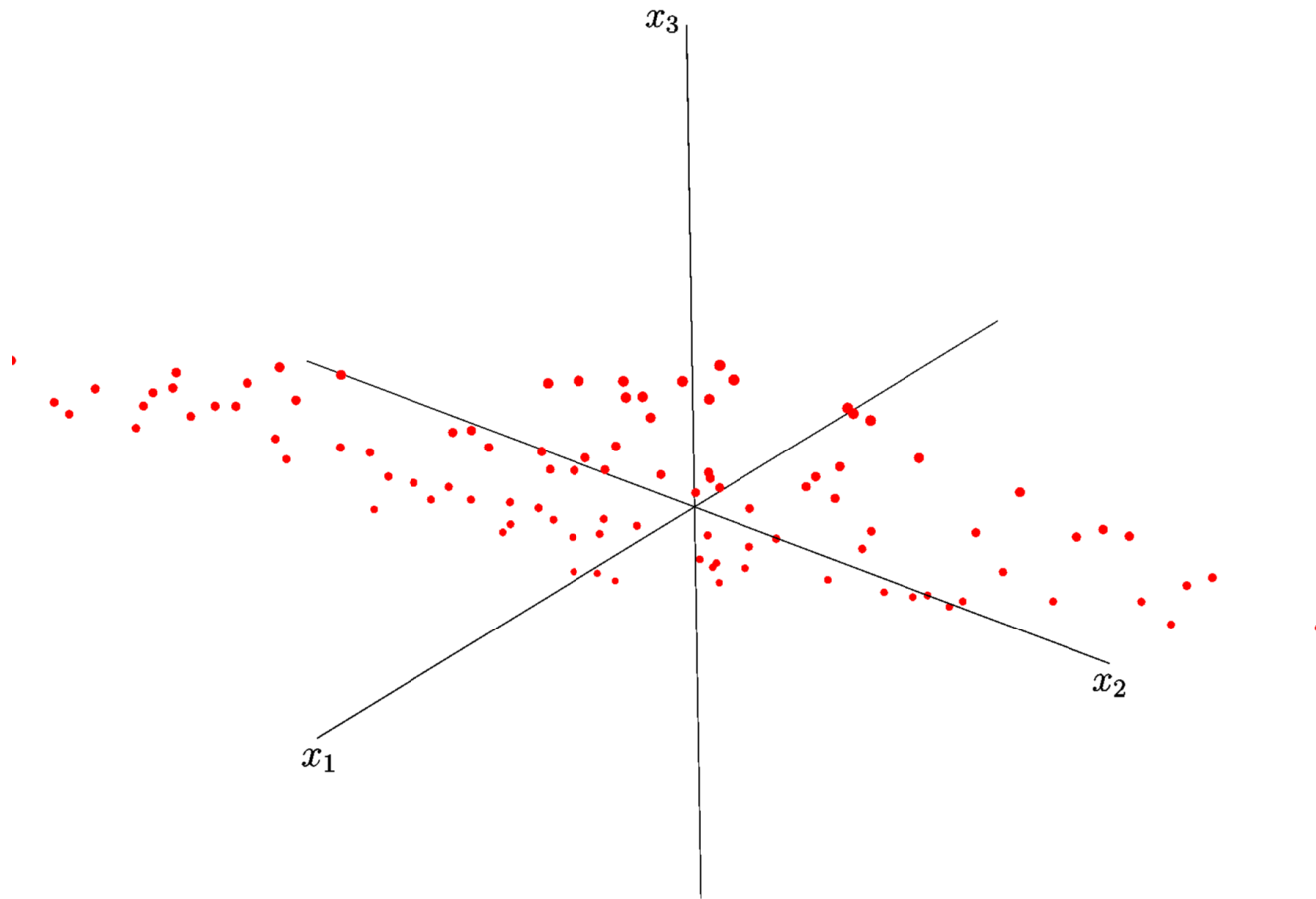
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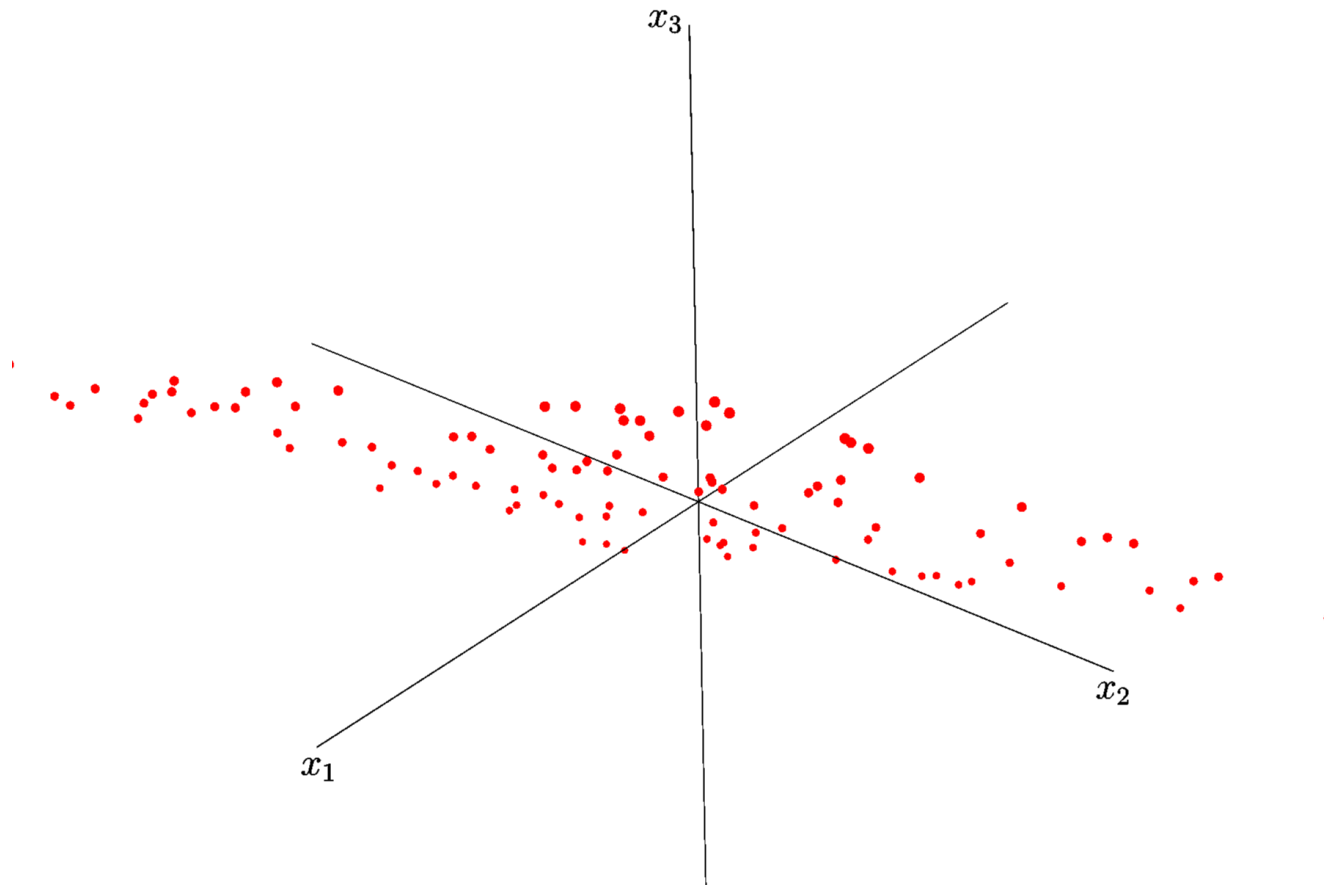
Can't spot low dimensional data by looking at numbers





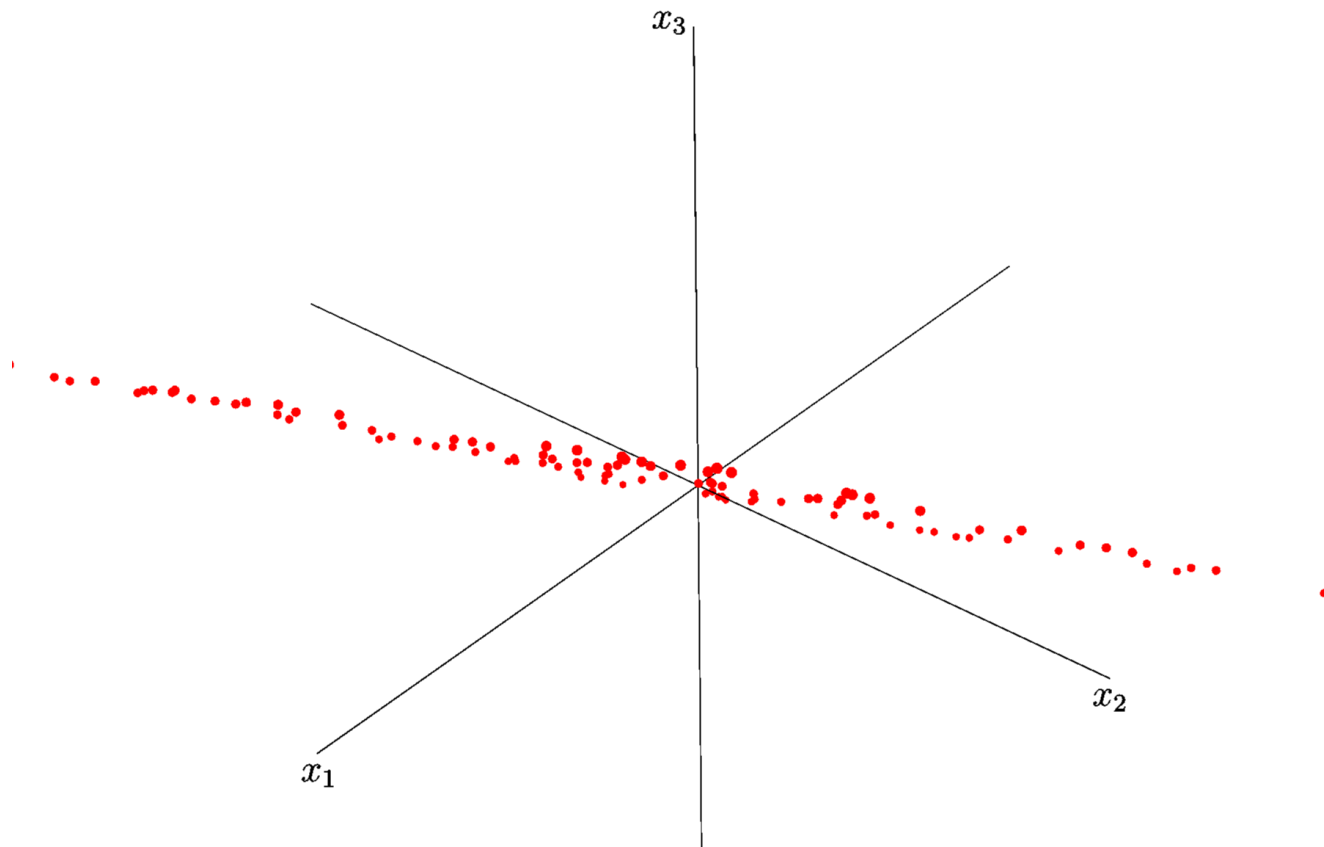
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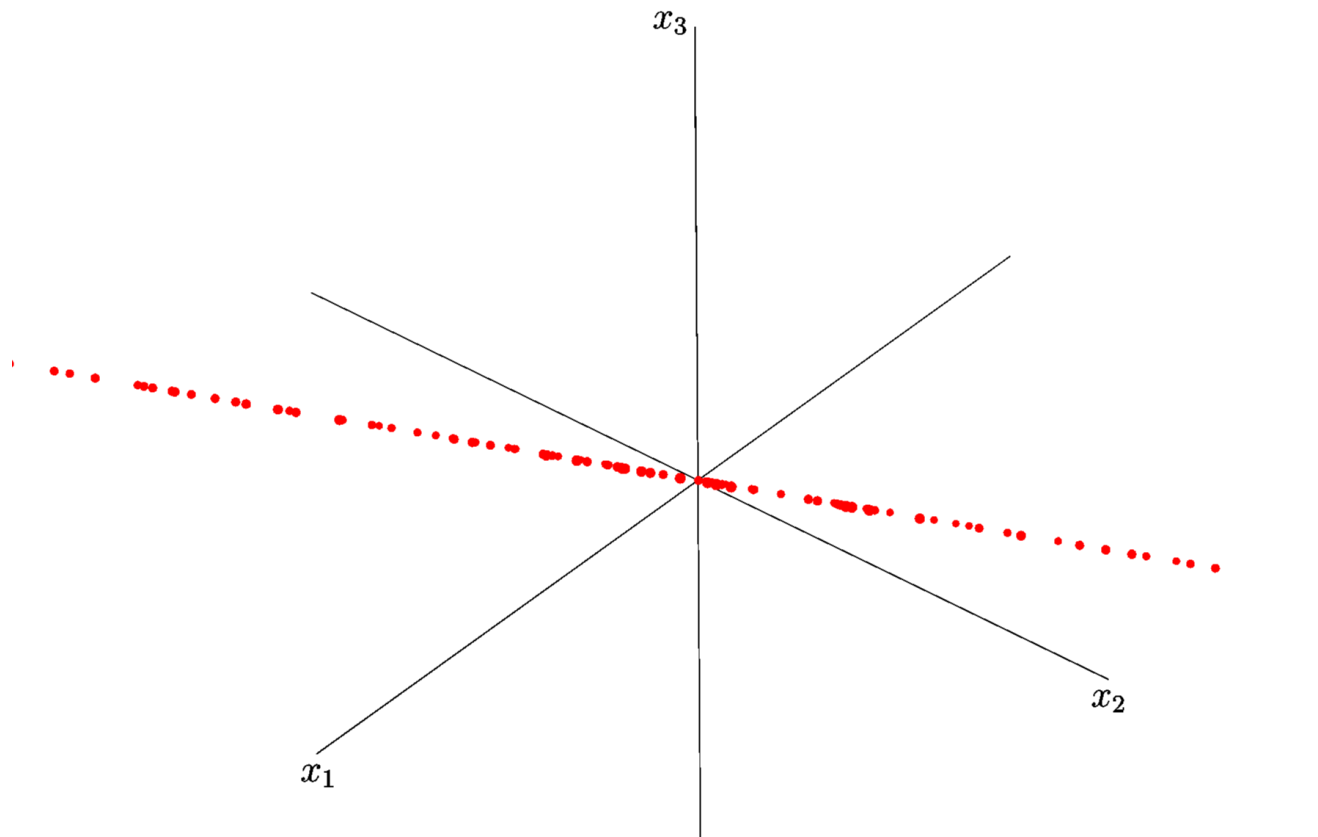
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# Dimensionality Reduction

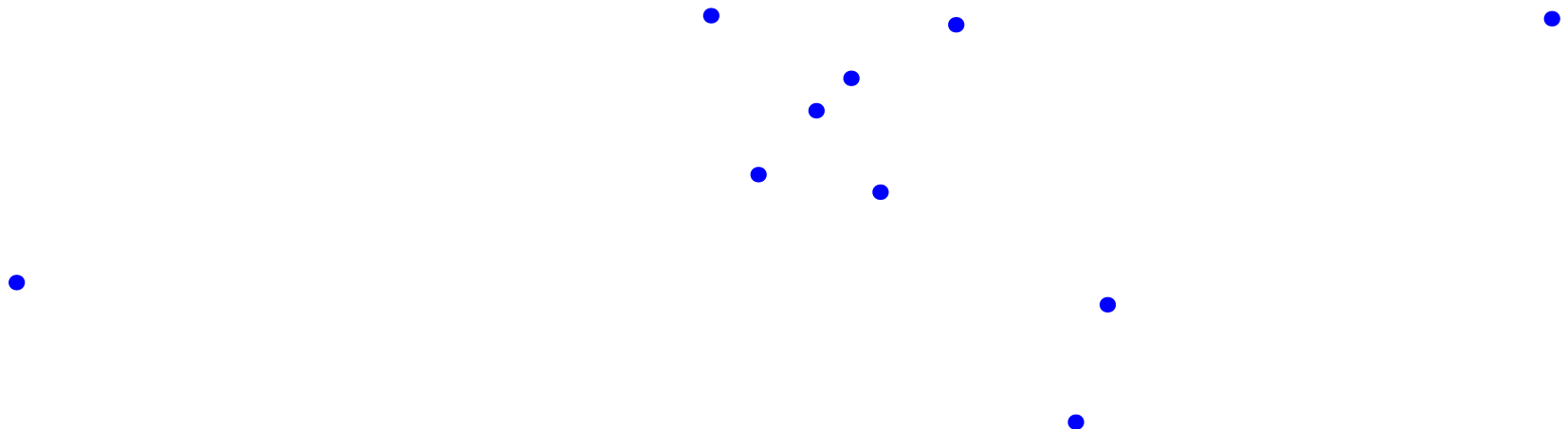
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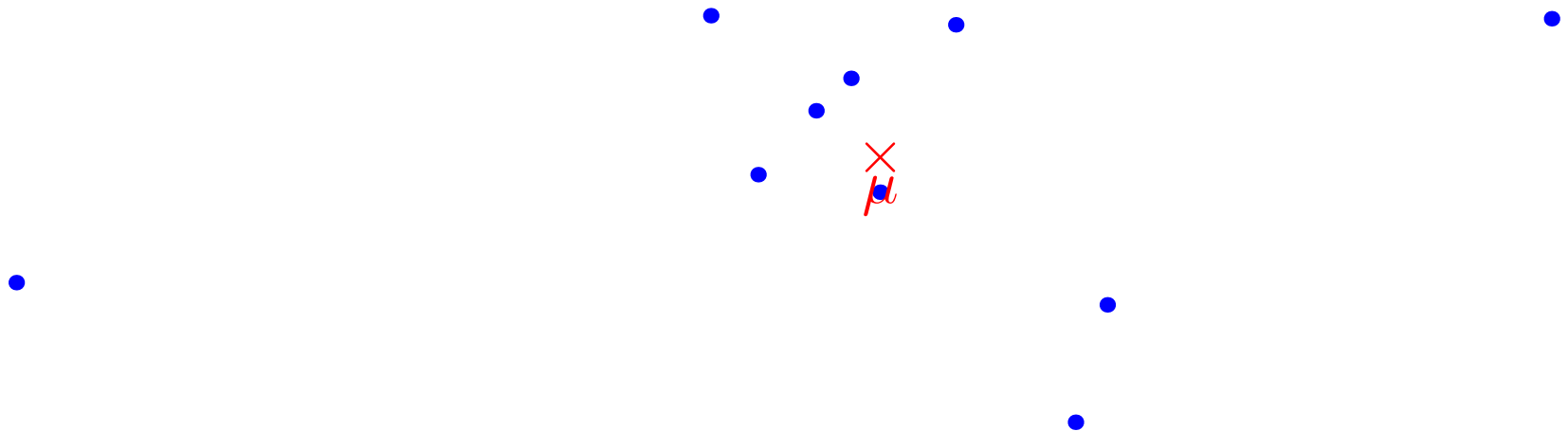
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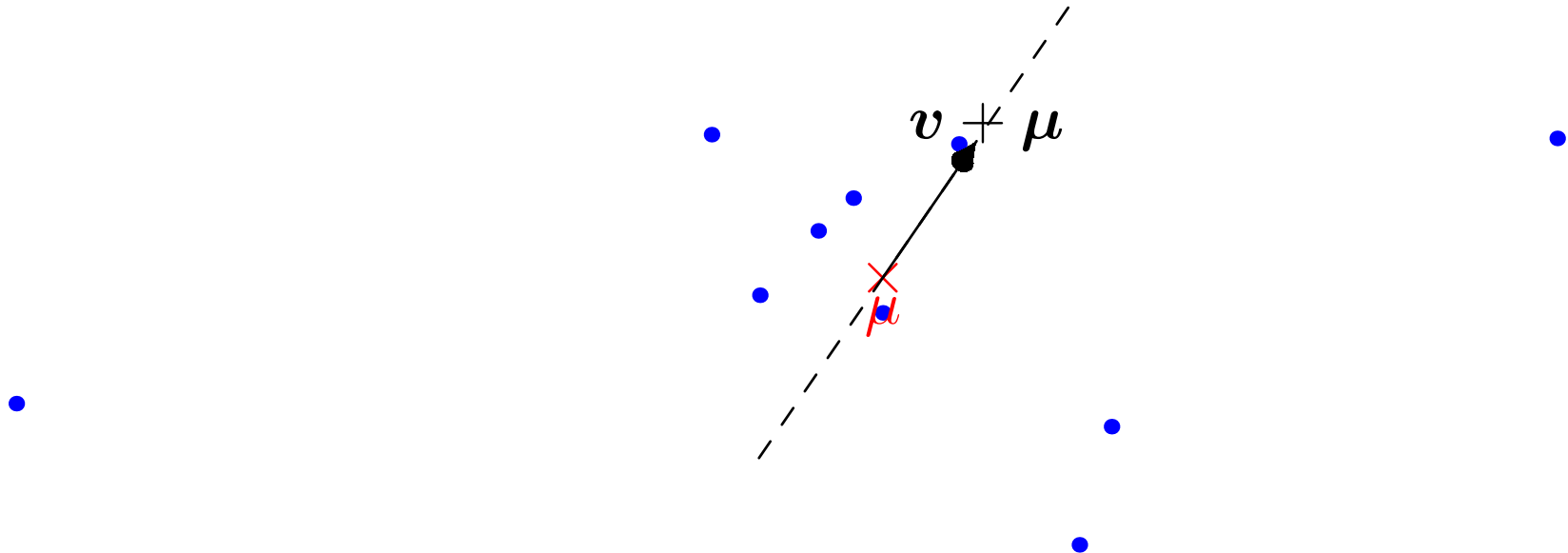
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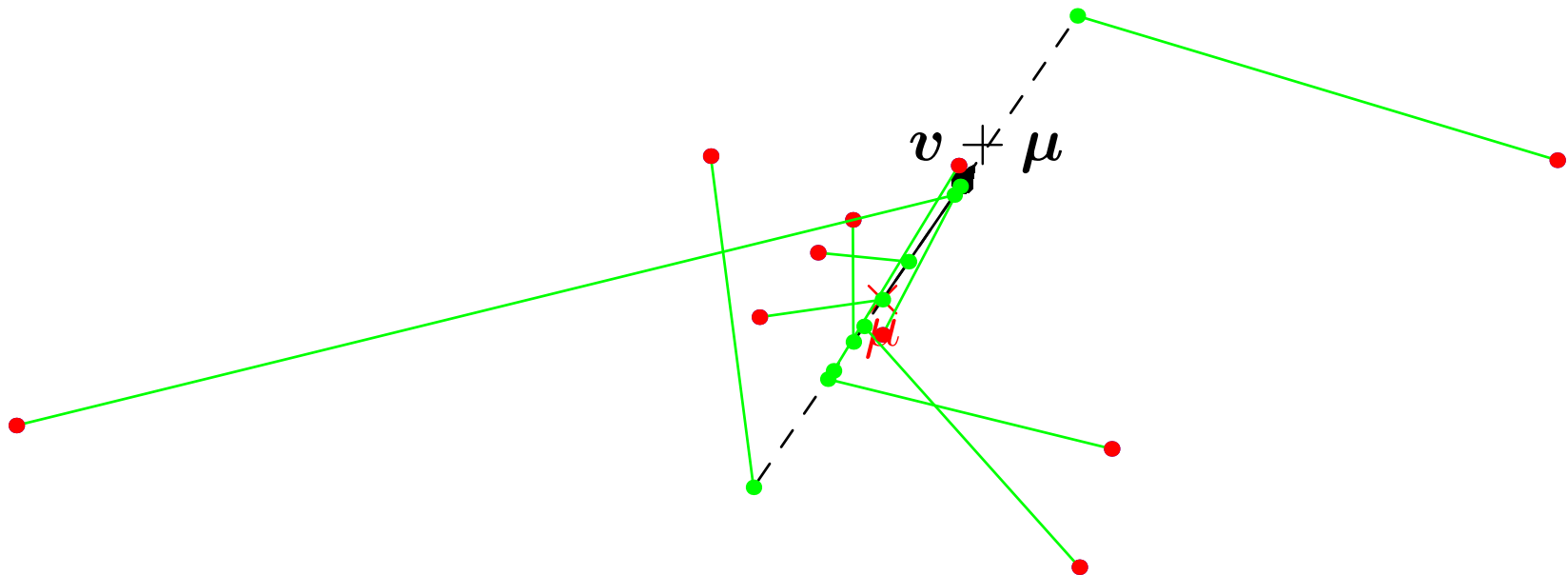
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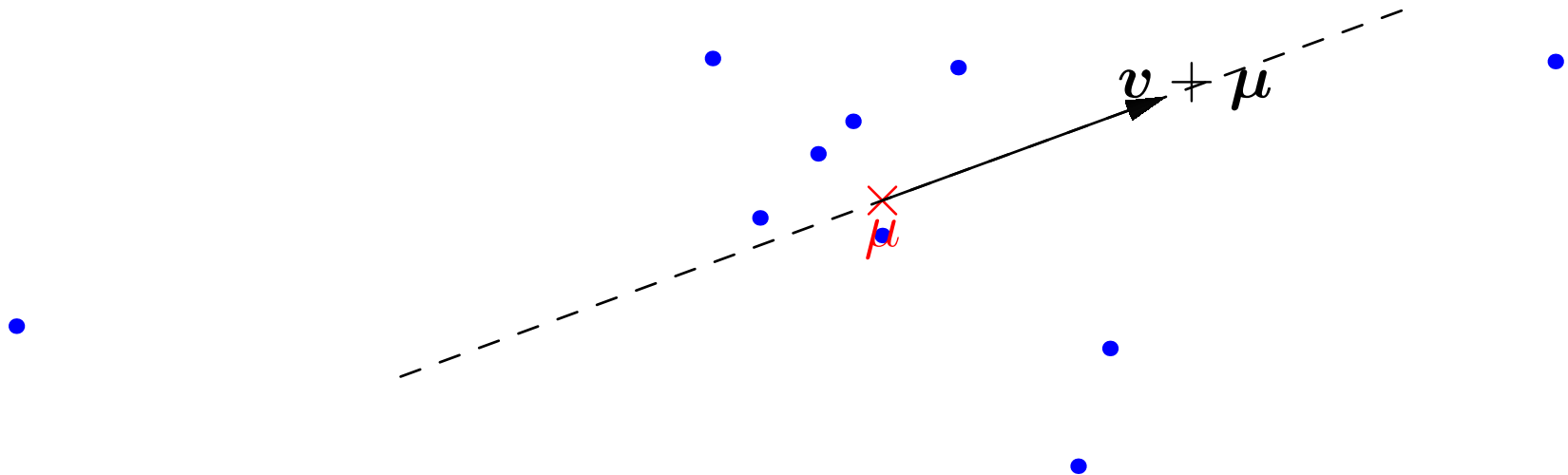
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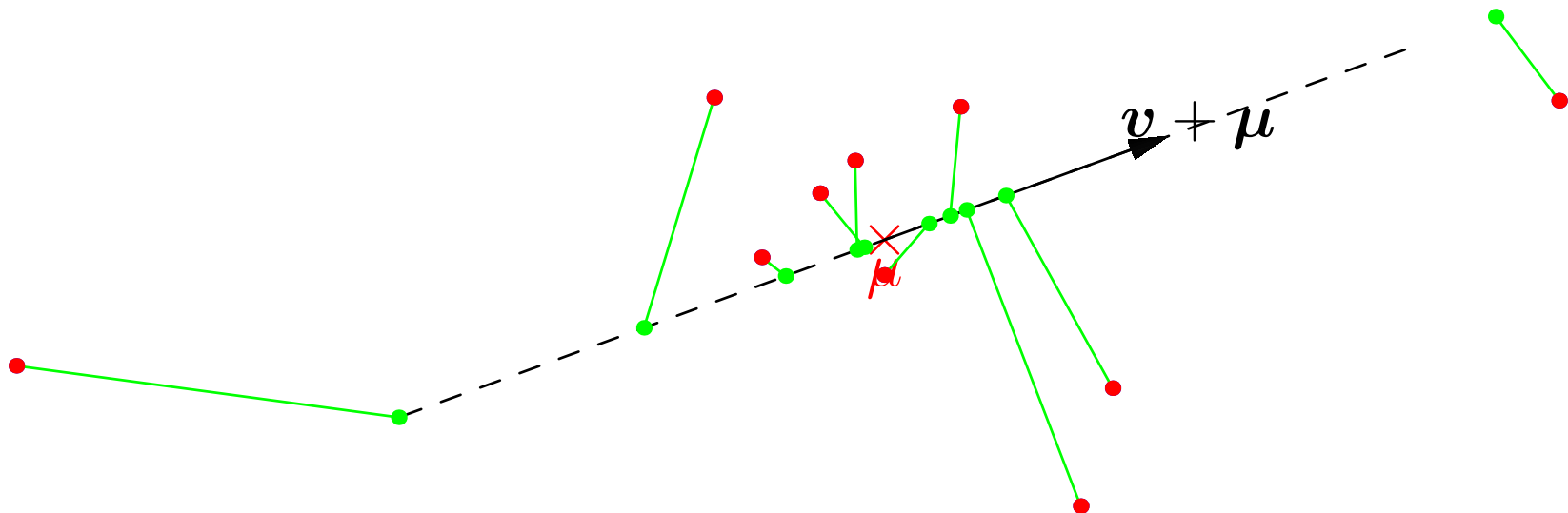
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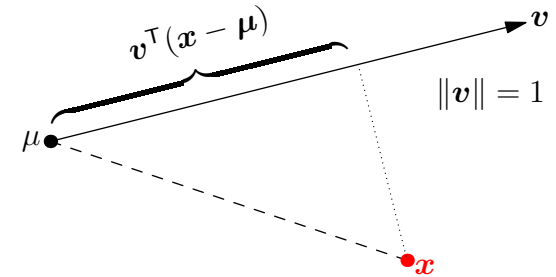
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# Direction of Maximum Variation

- Look for the vector  $\mathbf{v}$  with  $\|\mathbf{v}\|^2 = 1$  to maximise

$$\sigma^2 = \frac{1}{m-1} \sum_{i=1}^m (\mathbf{v}^\top (\mathbf{x}_i - \boldsymbol{\mu}))^2$$



- This is a constrained optimisation problem
- Solve by maximising Lagrangian

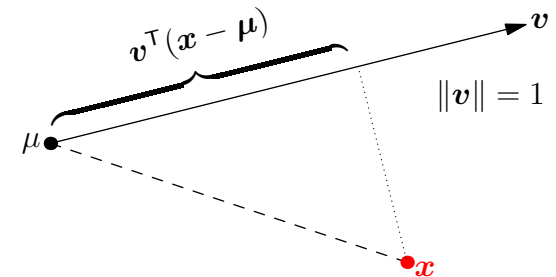
$$\mathcal{L} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{v}^\top (\mathbf{x}_k - \boldsymbol{\mu}))^2 - \lambda (\|\mathbf{v}\|^2 - 1)$$

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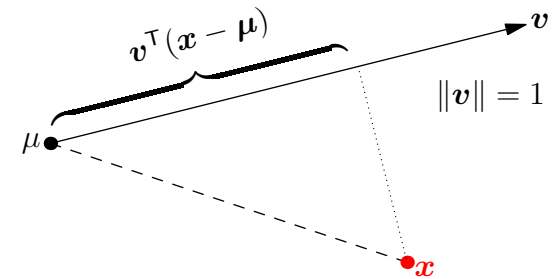
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$$\nabla \mathcal{L} = 2(\mathbf{C} \mathbf{v} - \lambda \mathbf{v}) = 0 \quad \Rightarrow \quad \mathbf{C} \mathbf{v} = \lambda \mathbf{v}$$

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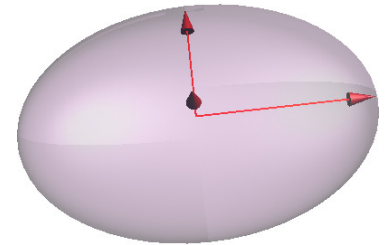
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# Direction of Maximum Variation

- The eigenvectors are directions that are extrema of the variance



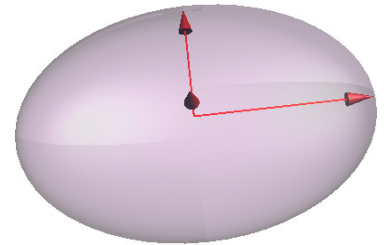
- The variance in direction  $\mathbf{v}$  is equal to

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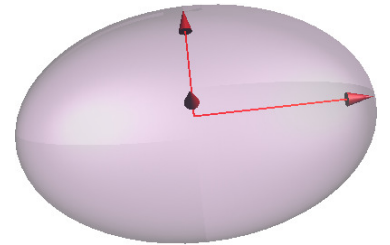
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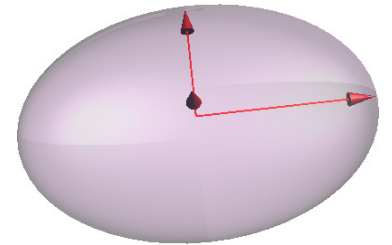
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# Covariance Matrix

- The **covariance matrix** is defined as

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^\top$$

- The components  $C_{ij}$  measure how the  $i^{th}$  and  $j^{th}$  components co-vary

$$C_{ij} = \frac{1}{m-1} \sum_{k=1}^m (x_{ik} - \mu_i)(x_{jk} - \mu_j)$$

- C.f. covariance of random variables

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

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- Remember that the outer-product of two vectors is defined as

$$\mathbf{x}\mathbf{y}^\top = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} (y_1 \quad y_2 \quad \cdots \quad y_n) = \begin{pmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & x_n y_2 & \cdots & x_n y_n \end{pmatrix}$$

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# Matrix Form

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$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^\top$$

- Define the matrix

$$\mathbf{X} = \frac{1}{\sqrt{m-1}} (\mathbf{x}_1 - \boldsymbol{\mu}, \mathbf{x}_2 - \boldsymbol{\mu}, \dots, \mathbf{x}_m - \boldsymbol{\mu})$$

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# Properties of Covariance Matrix

- The **quadratic form** of a vector and matrix is defined as

$$\mathbf{v}^T \mathbf{M} \mathbf{v}$$

- The quadratic form of a covariance matrix is non-negative for any vector

$$\mathbf{v}^T \mathbf{C} \mathbf{v} = \mathbf{v}^T \mathbf{X} \mathbf{X}^T \mathbf{v} = \mathbf{u}^T \mathbf{u} = \|\mathbf{u}\|^2 \geq 0$$

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# Surface Defined by Matrix

- The set of vectors  $x$  such that

$$x^T C^{-1} x = 1$$

defines a surface

- The surface is an ellipsoid,  $\mathcal{E}$
- The eigenvectors point in the direction of the principal axes of the ellipsoid
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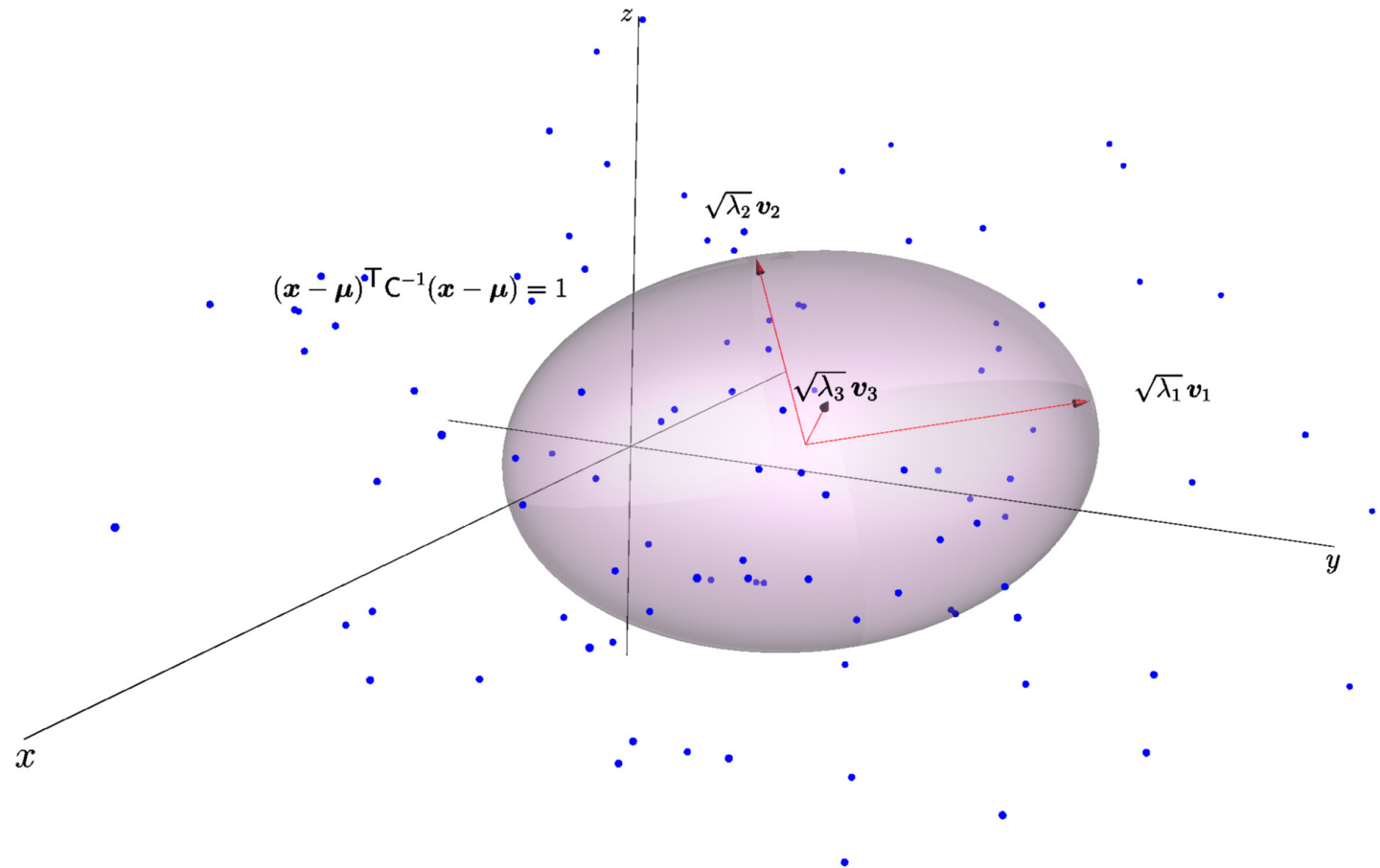
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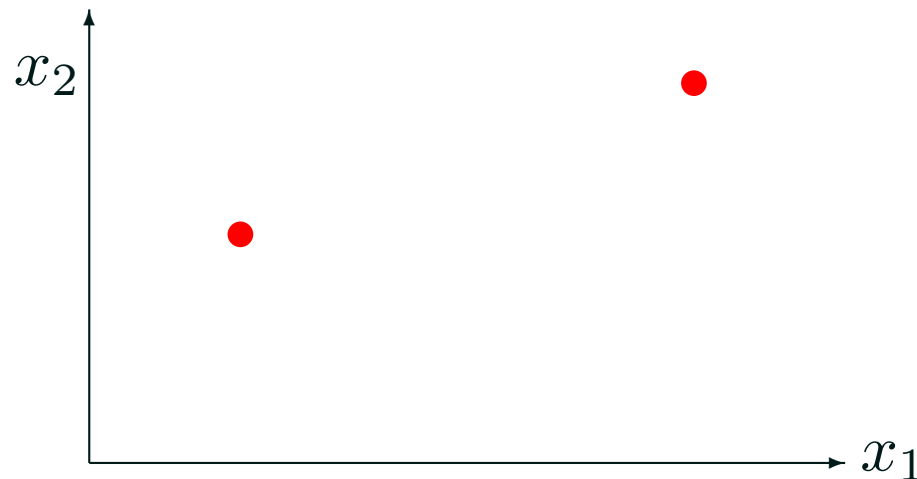
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# Ellipsoid and Eigen Space



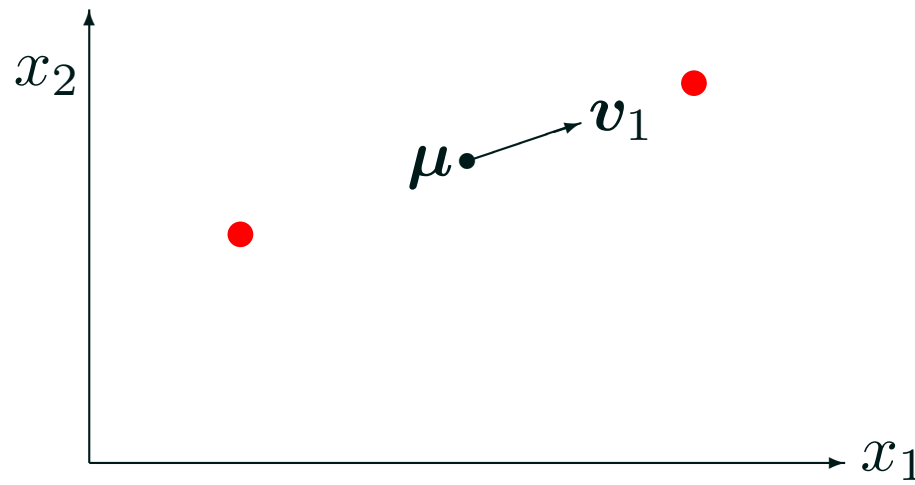
# Spanning Input Space

- A covariance matrix will have a zero eigenvalue only if there is no variation in the direction of the corresponding eigenvector
- A covariance matrix will have zero eigenvalues if the number of patterns are less than or equal to the number of dimensions



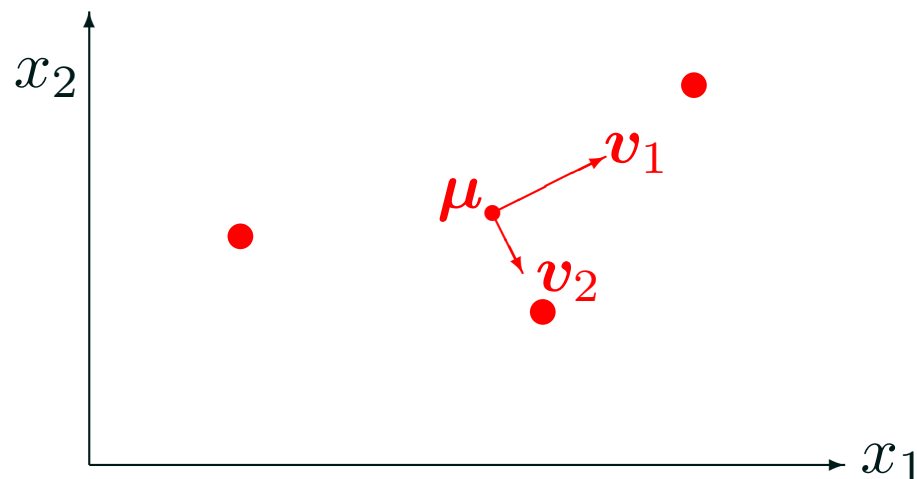
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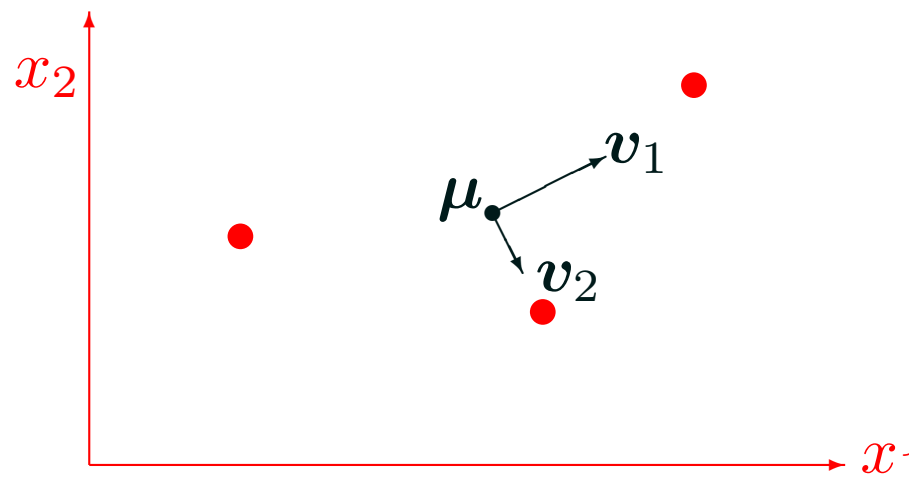
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# Positive Definite

- Matrices with no zero eigenvalues are called **full rank** matrices (as opposed to rank deficient)
- Full rank matrices are invertible, rank deficient matrices are singular and non-invertible
- Full rank covariance matrices that have positive eigenvalues only and are said to be **positive definite**
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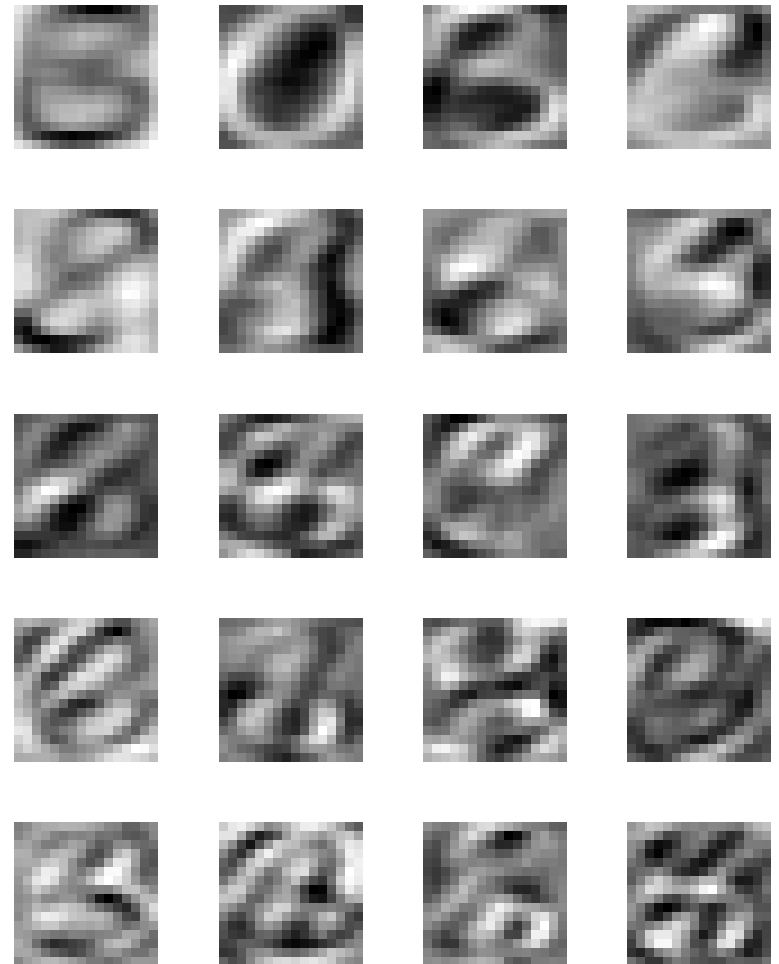
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# Outline

1. Covariance Matrices
2. **Principal Component Analysis**
3. Duality



# Principal Component Analysis

- PCA occurs as follows
  - ★ Construct the covariance matrix
  - ★ Find the eigenvalues and eigenvectors
  - ★ Keep the eigenvectors with the largest eigenvalues (principal components)
  - ★ Project the inputs into the space spanned by the principal components
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# Projection Matrix

- To project the inputs construct the projection matrix

$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \end{pmatrix}$$

- $k < p$  is the number of principal components we keep
- Given a  $p$ -dimensional input pattern  $\mathbf{x}$  we can construct a  $k$ -dimensional representation  $\mathbf{z}$

$$\mathbf{z} = \mathbf{P}(\mathbf{x} - \boldsymbol{\mu})$$

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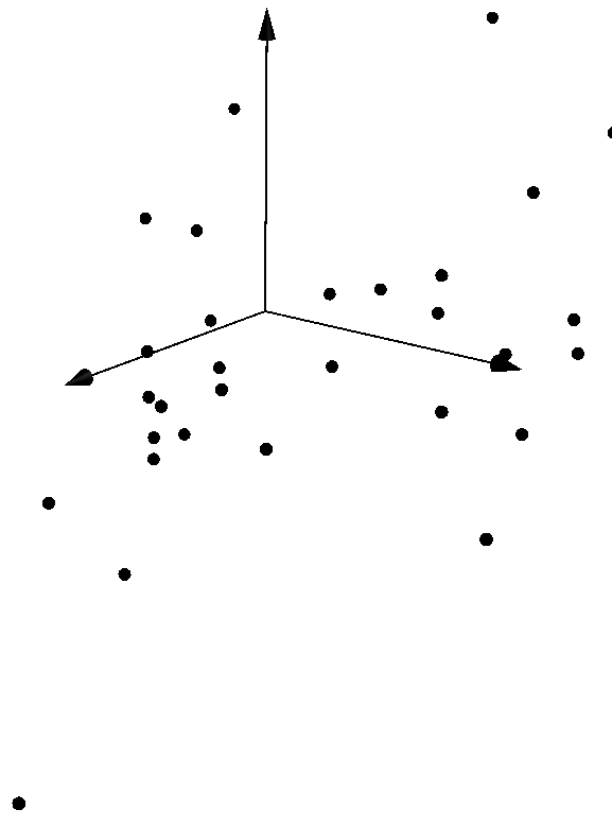
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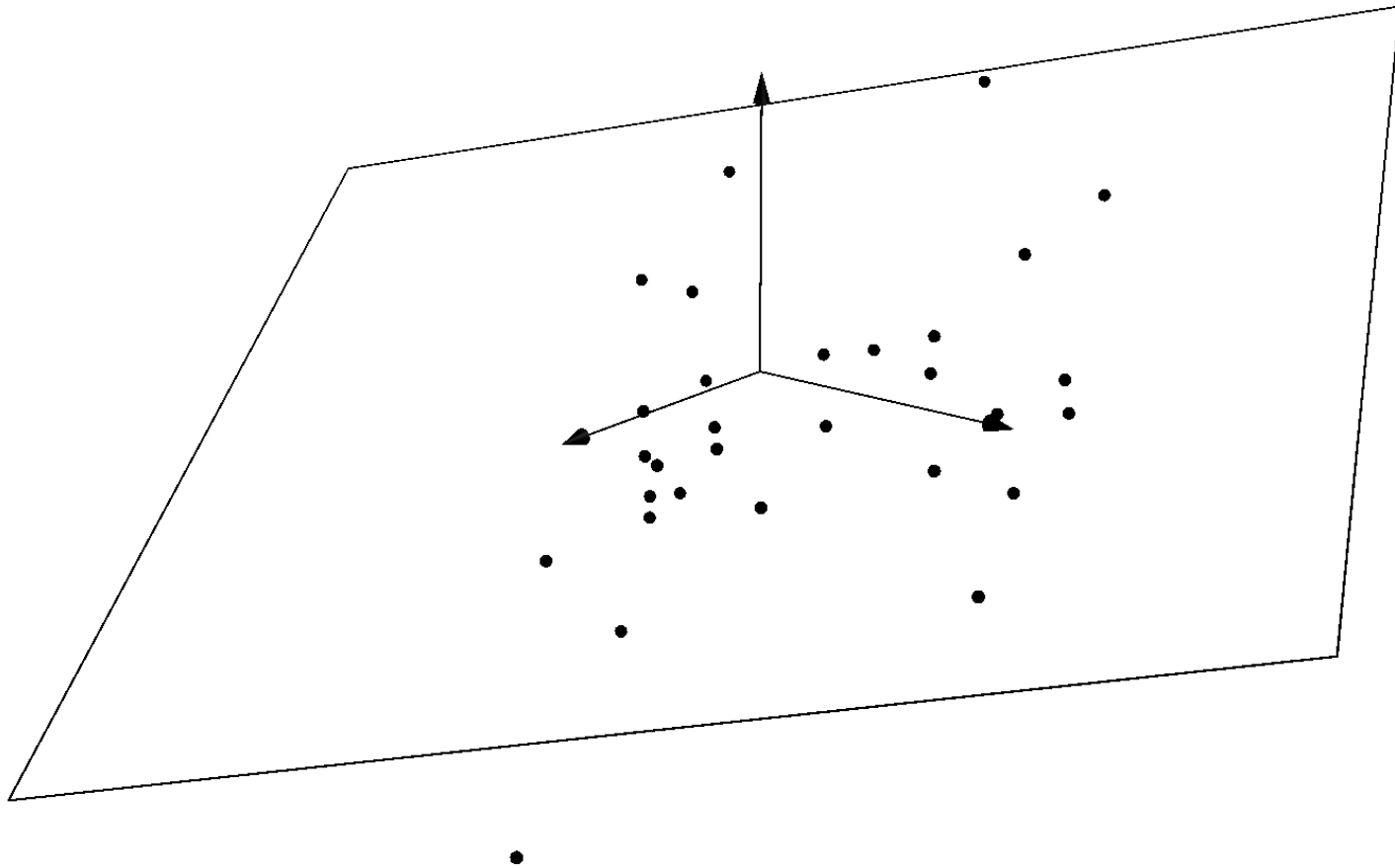
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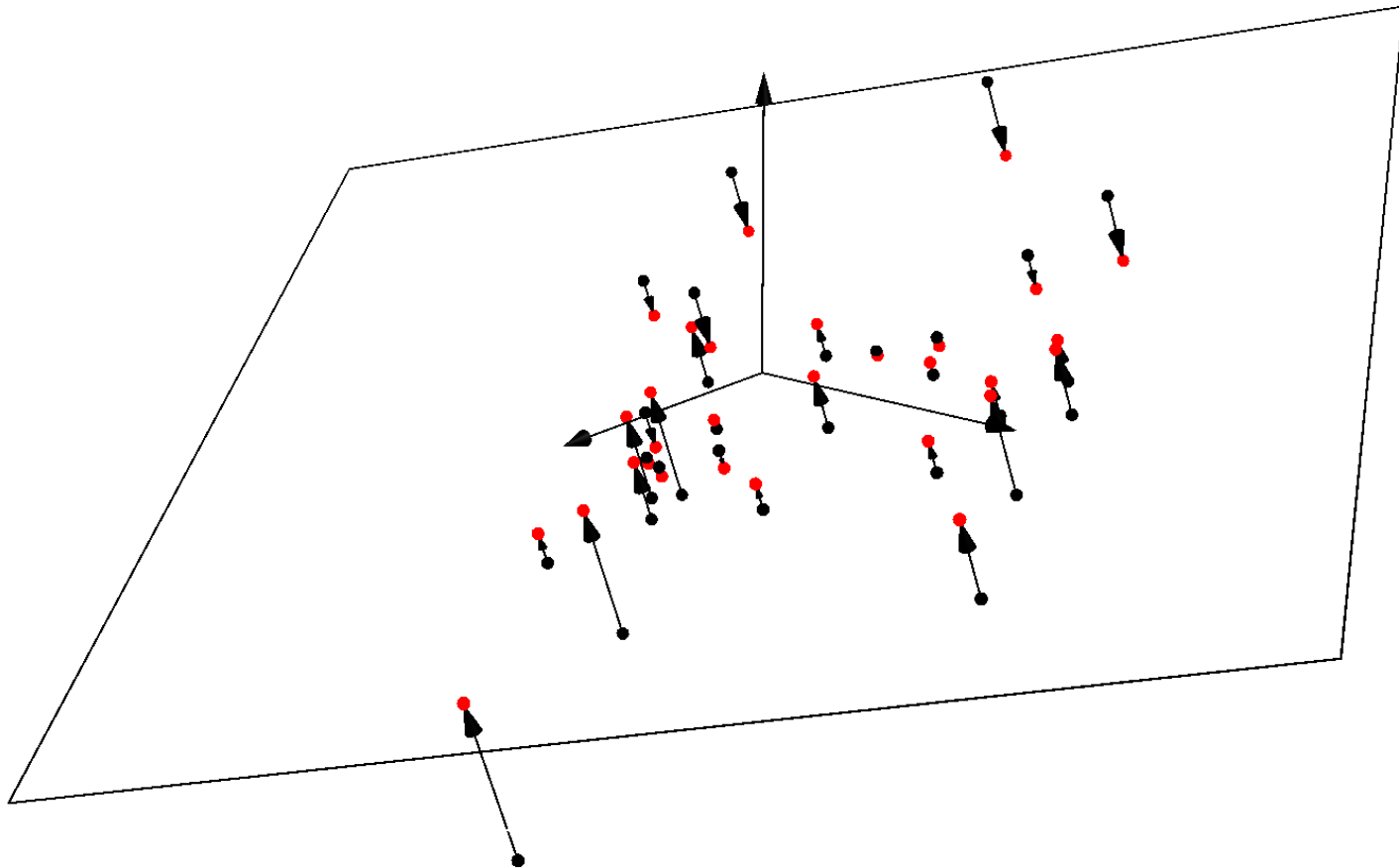
# Subspace Projection



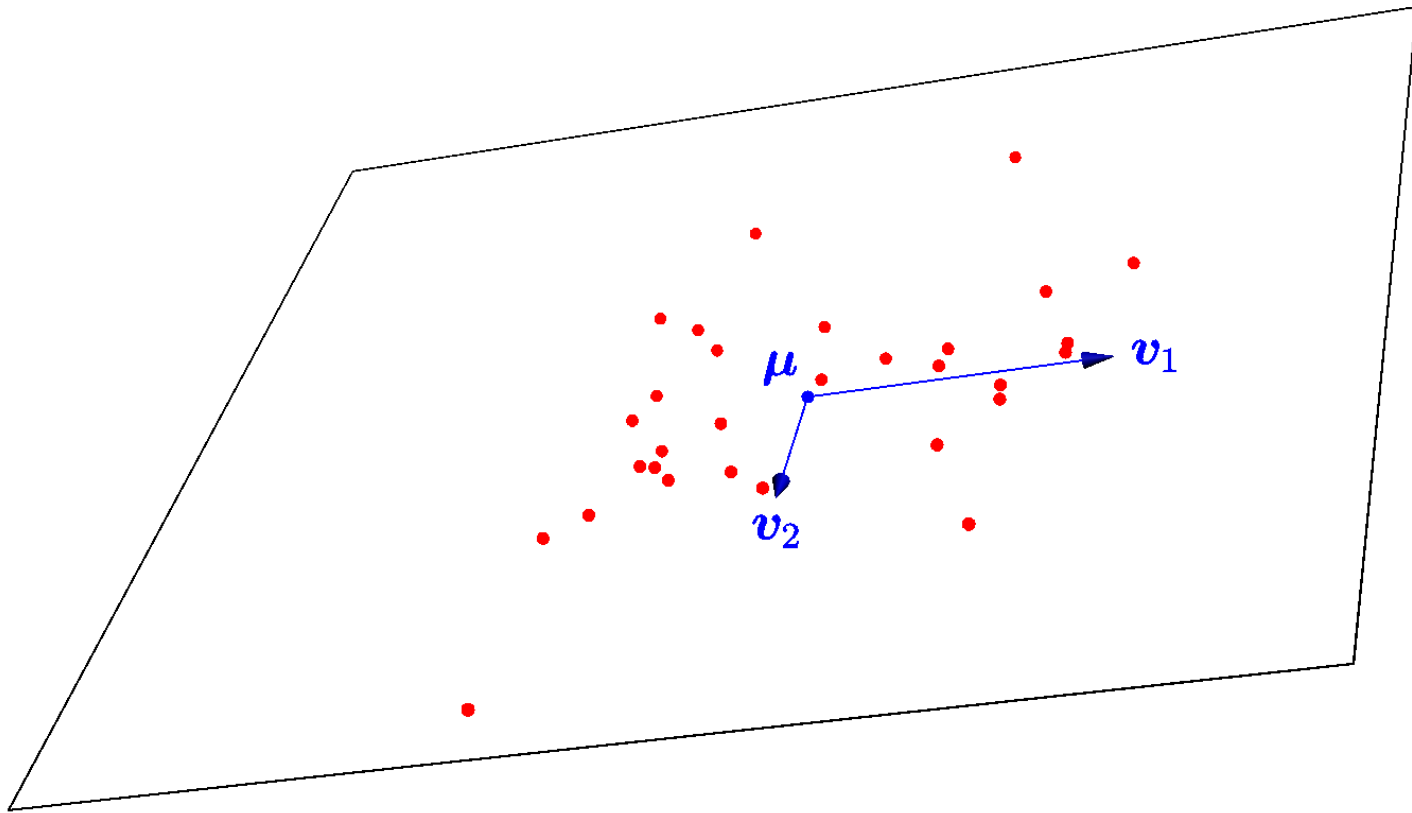
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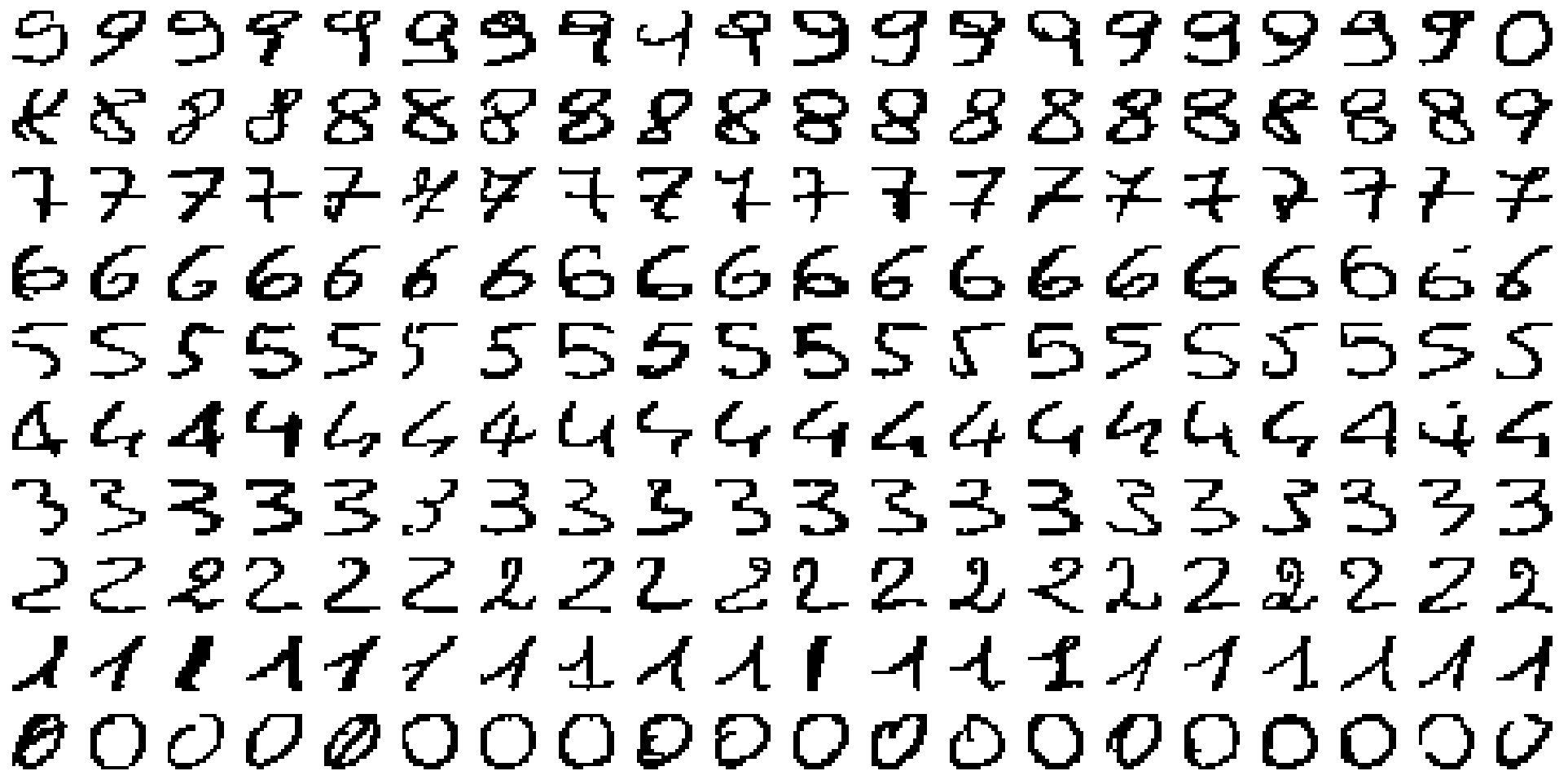
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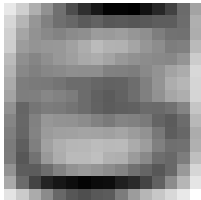
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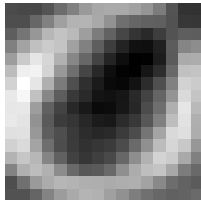
# Hand Written Digits



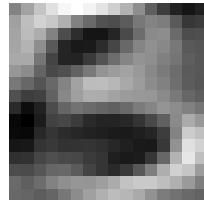
# Eigenvectors



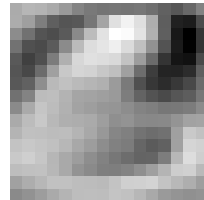
$\mu$



$v_1$



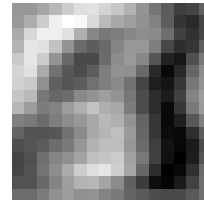
$v_2$



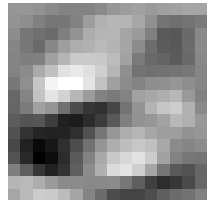
$v_3$



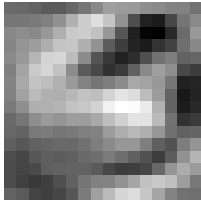
$v_4$



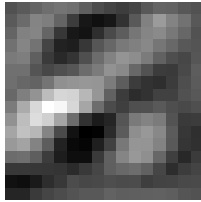
$v_5$



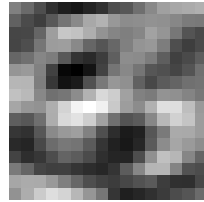
$v_6$



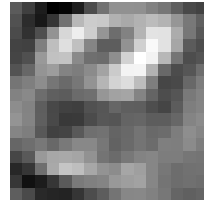
$v_7$



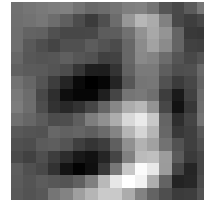
$v_8$



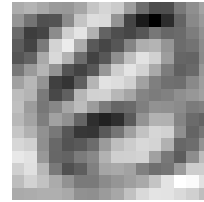
$v_9$



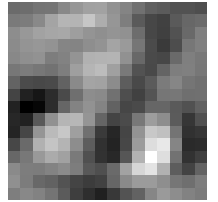
$v_{10}$



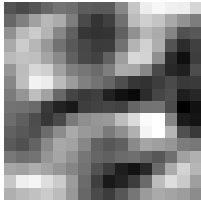
$v_{11}$



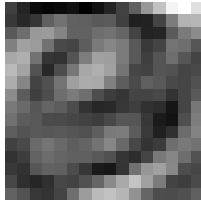
$v_{12}$



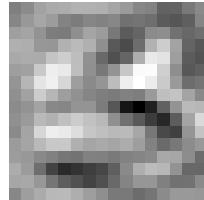
$v_{13}$



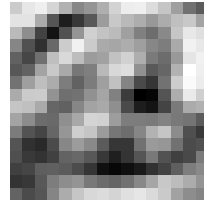
$v_{14}$



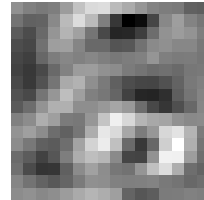
$v_{15}$



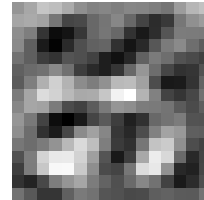
$v_{16}$



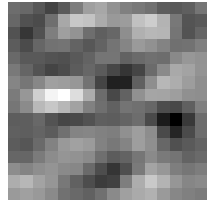
$v_{17}$



$v_{18}$



$v_{19}$



$v_{20}$

# Reconstruction

- Projecting into a subspace of eigenvectors can be seen as approximating the inputs by

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- Principle component analysis projects the data into a subspace of size  $m$  with the minimal approximation error  $\mathbb{E} [\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|^2]$
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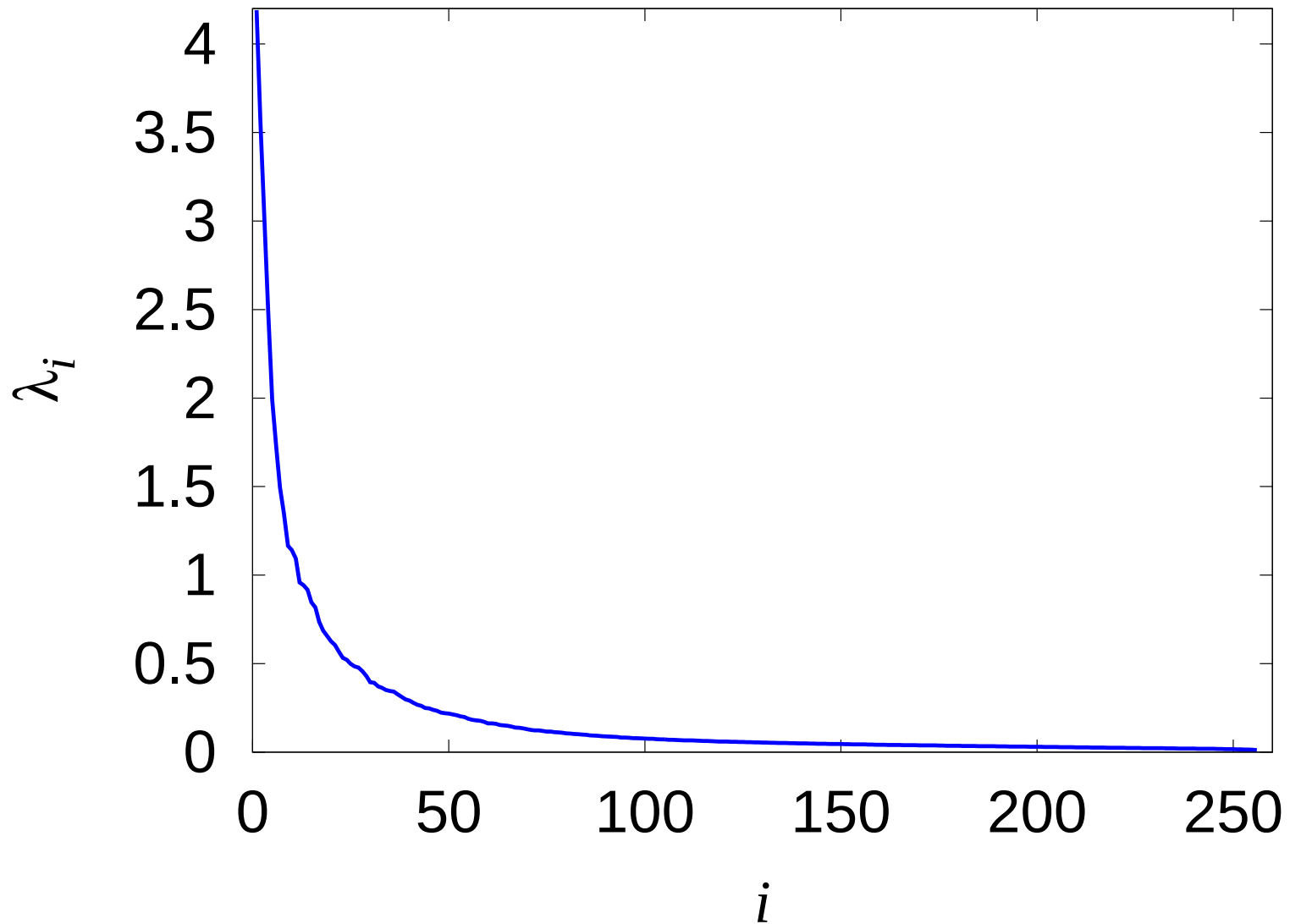
# Reconstruction

- Projecting into a subspace of eigenvectors can be seen as approximating the inputs by

$$\hat{\mathbf{x}}_i = \boldsymbol{\mu} + \sum_{j=1}^k z_j^i \mathbf{v}_j, \quad z_j^i = \mathbf{v}_j^\top (\mathbf{x}_i - \boldsymbol{\mu}), \quad \|\mathbf{v}_j\| = 1$$

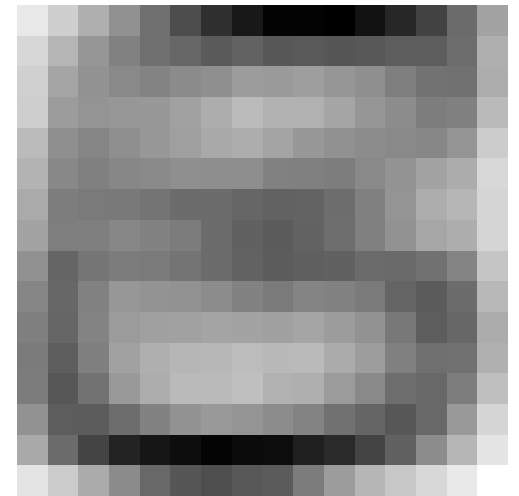
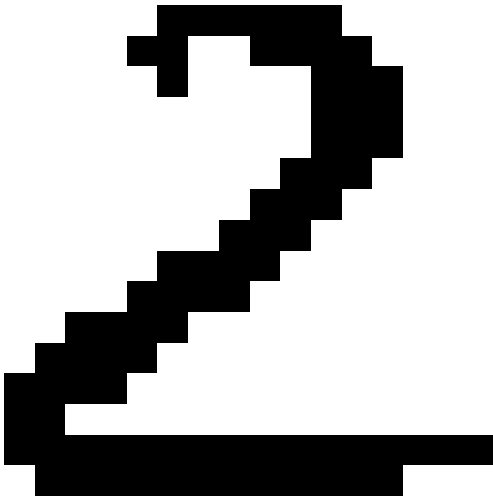
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# Eigenvalues for Digits



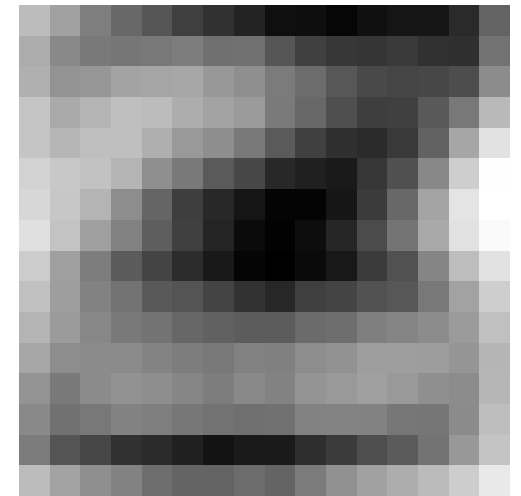
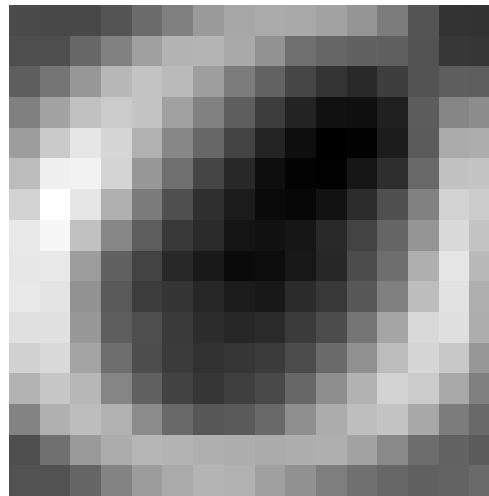
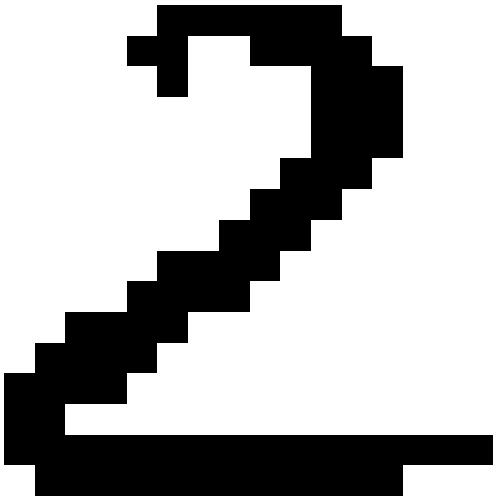
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1.6 -1.1 -1.6 2.1 -0.52 2.8 0.72 0.7 -0.68 -0.41 -1.4 -1.5 -0.54 -0.62 1.3 -1.4 -0.27 0.74 0.77 -1



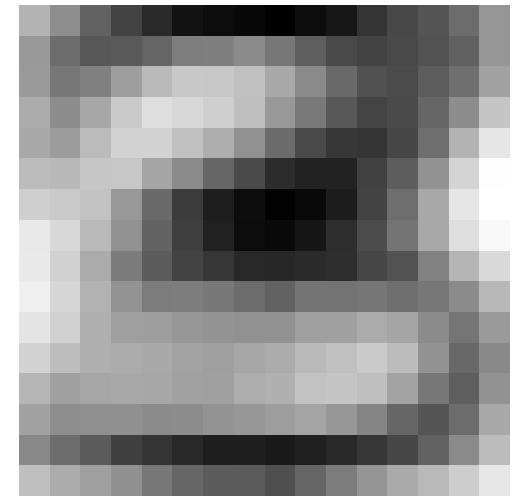
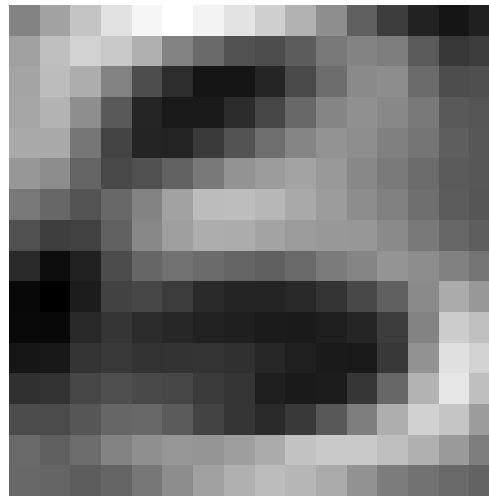
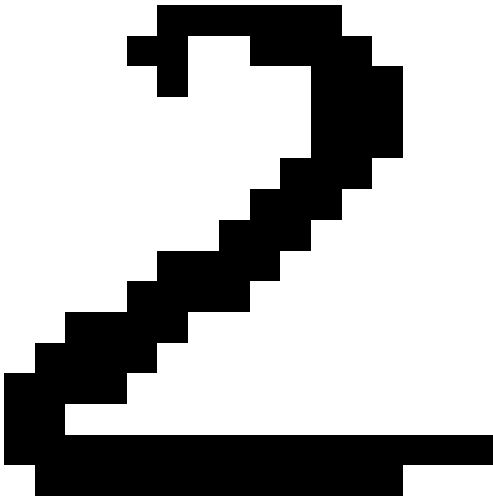
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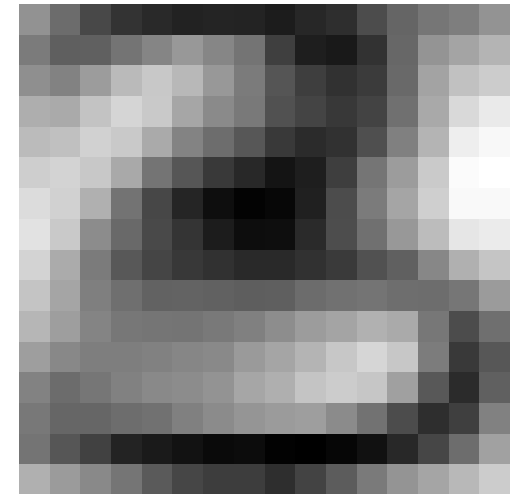
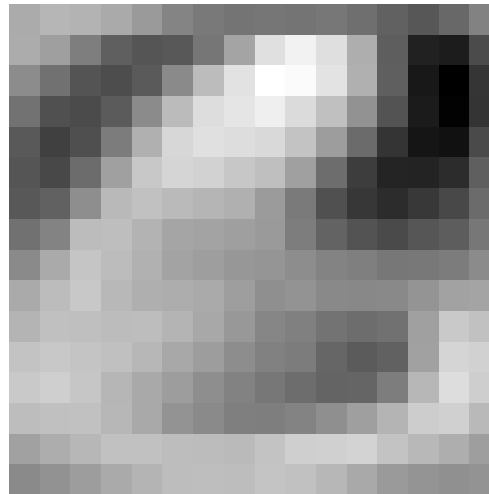
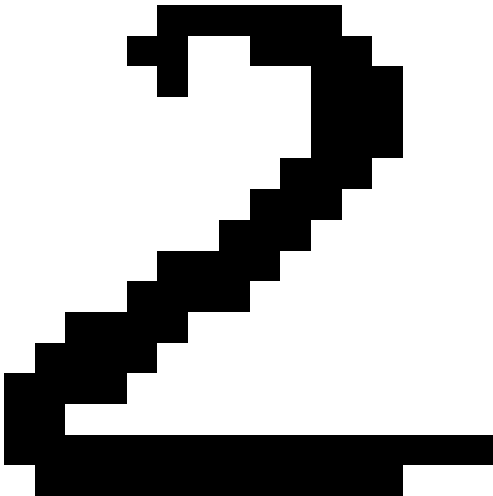
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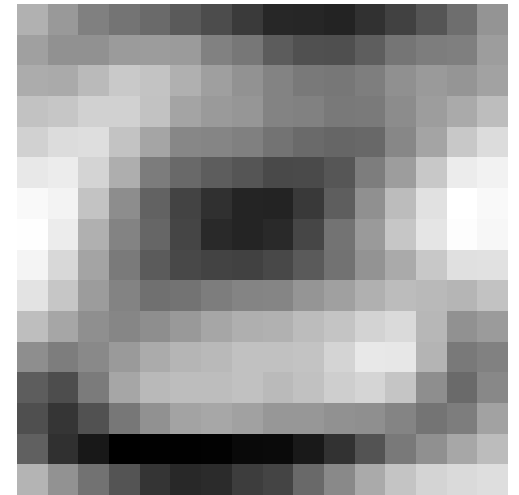
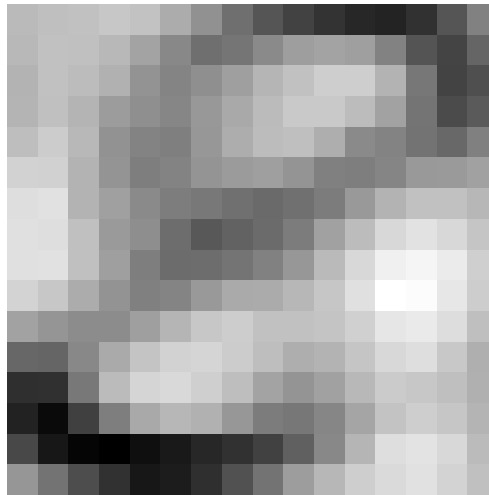
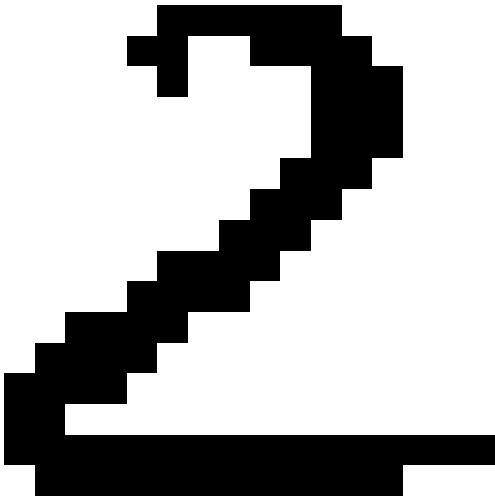
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# Reconstruction from Eigenvectors

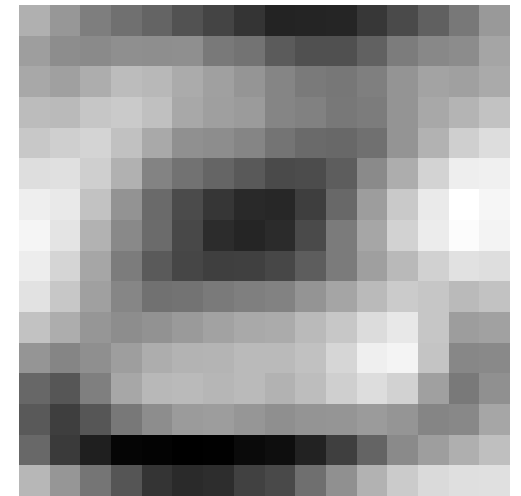
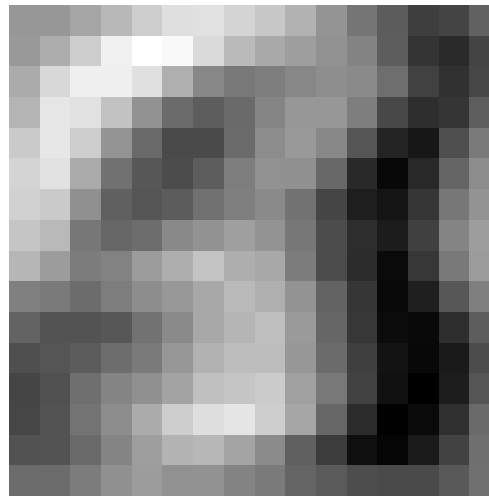
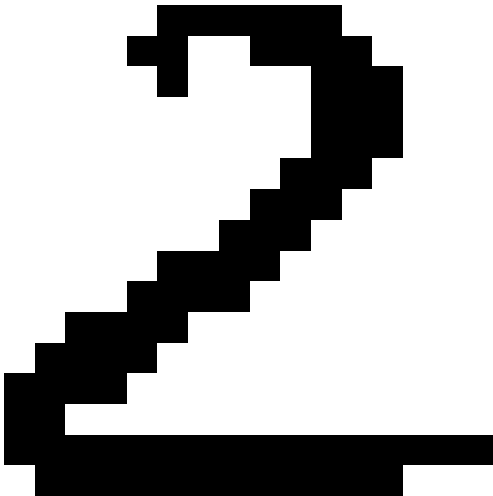
1.6 -1.1 -1.6 **2.1** -0.52 2.8 0.72 0.7 -0.68 -0.41 -1.4 -1.5 -0.54 -0.62 1.3 -1.4 -0.27 0.74 0.77 -1





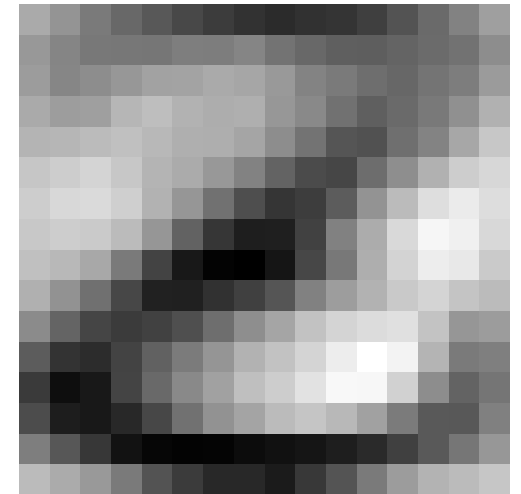
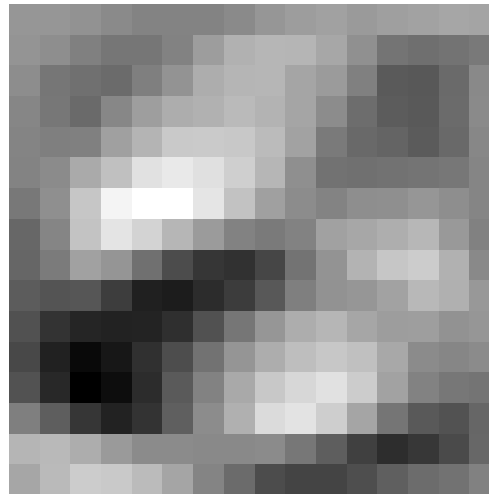
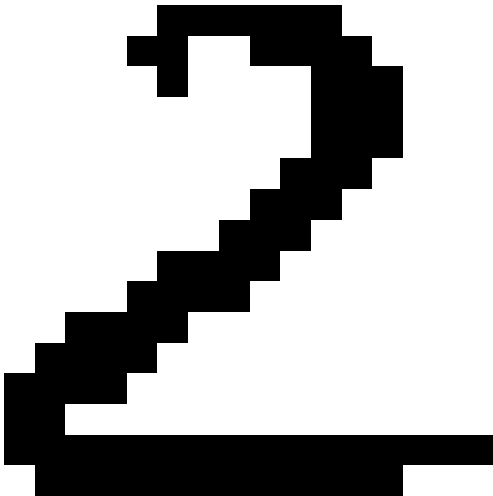
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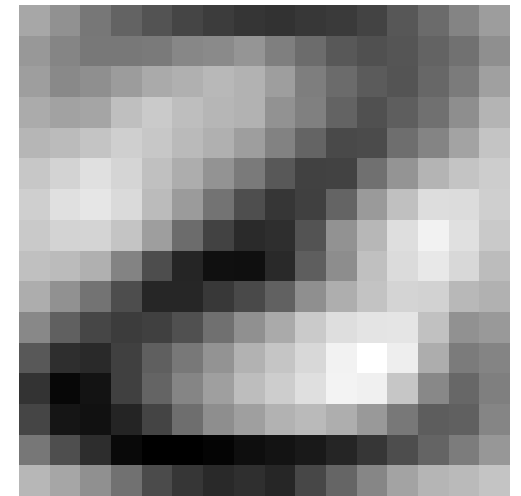
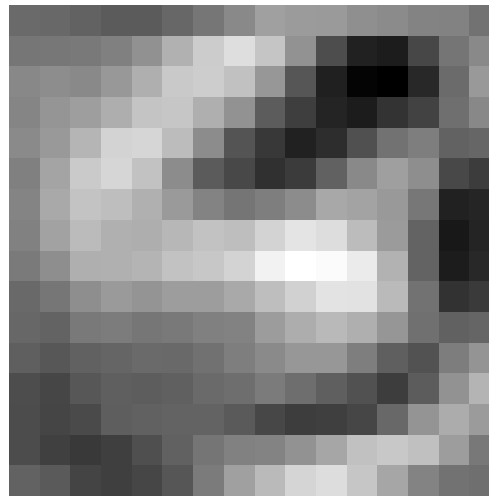
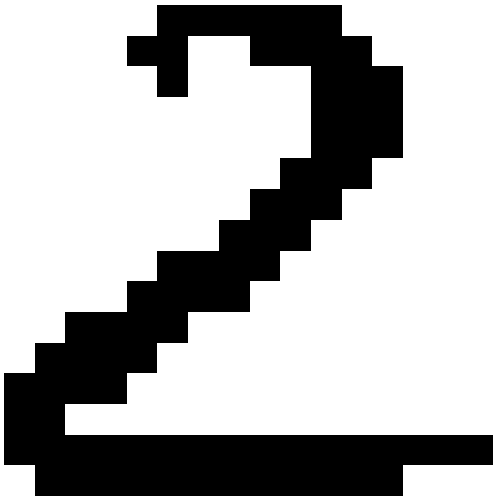
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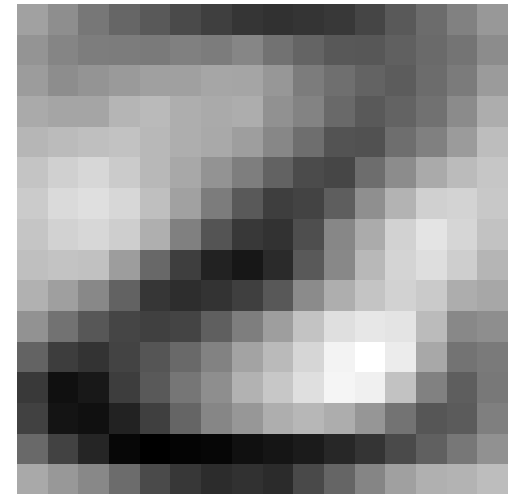
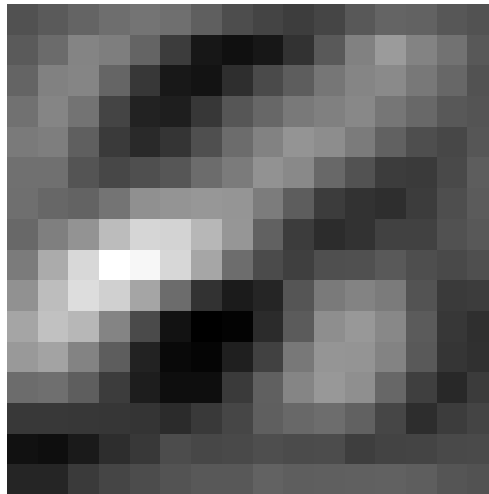
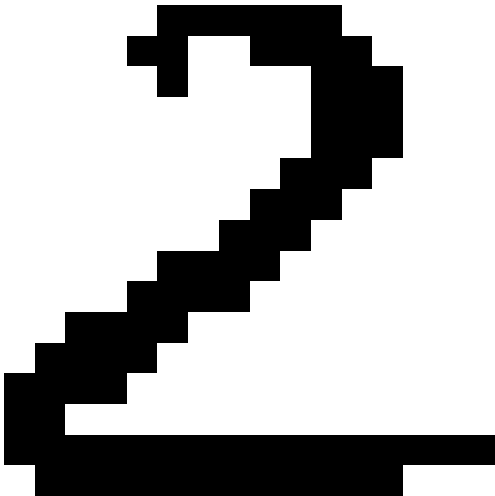
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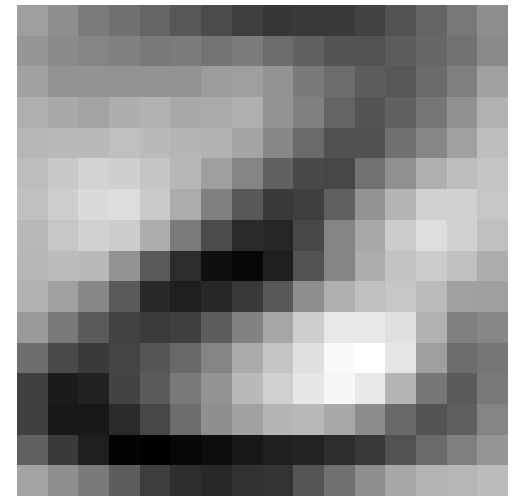
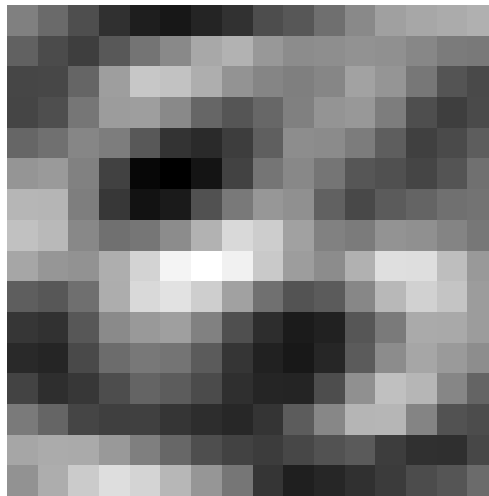
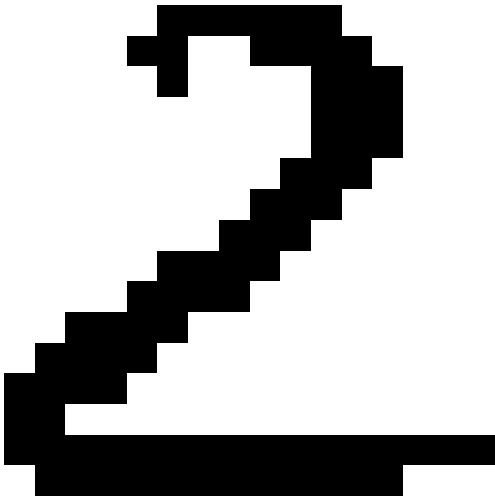
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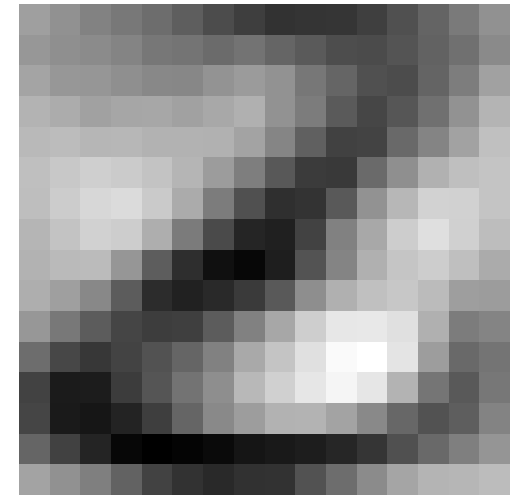
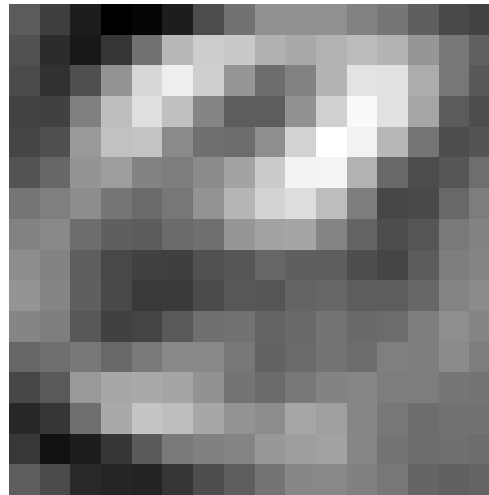
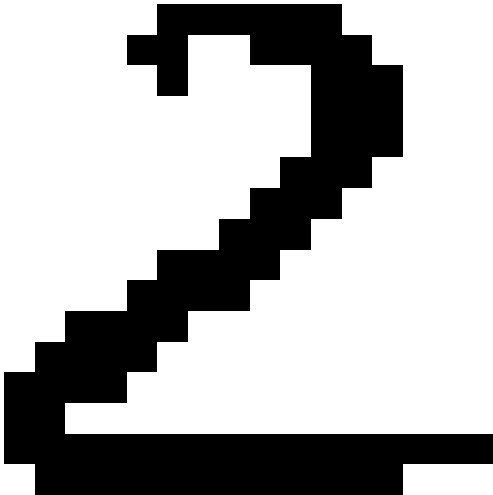
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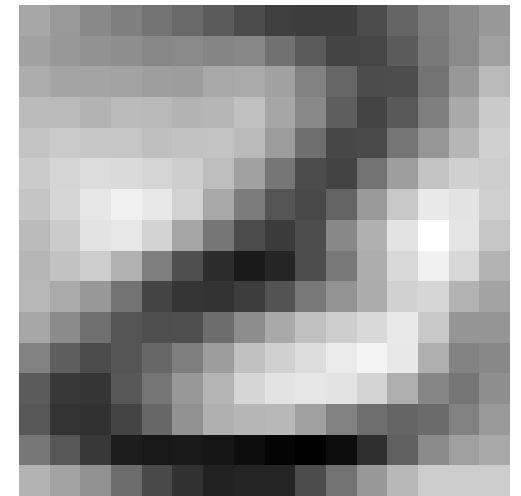
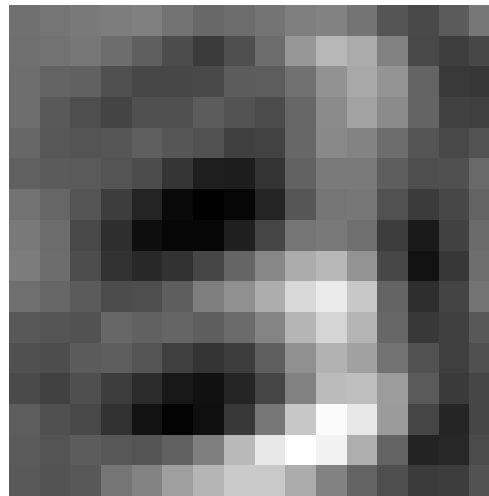
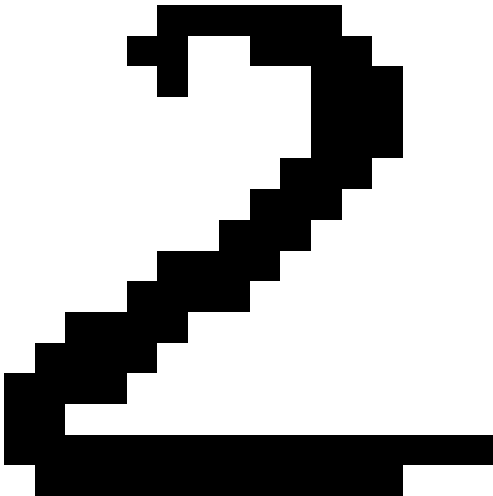
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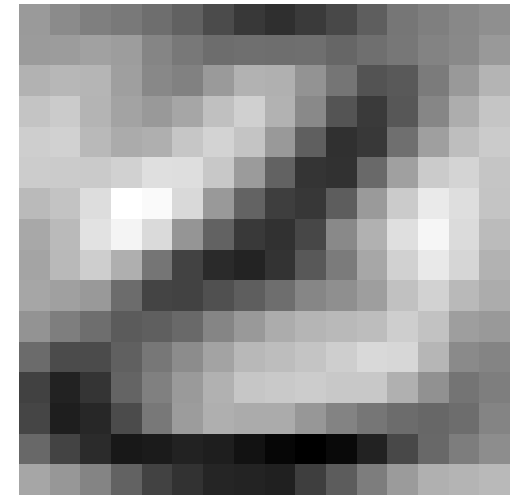
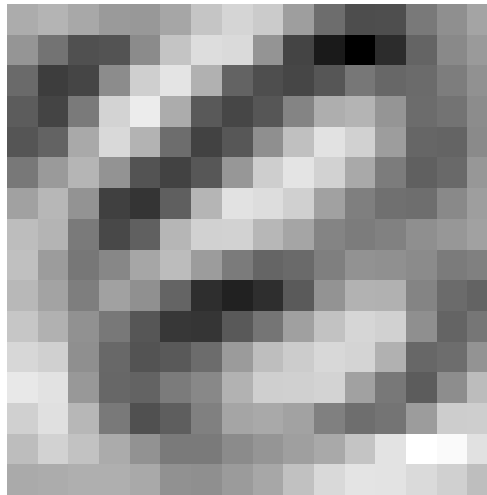
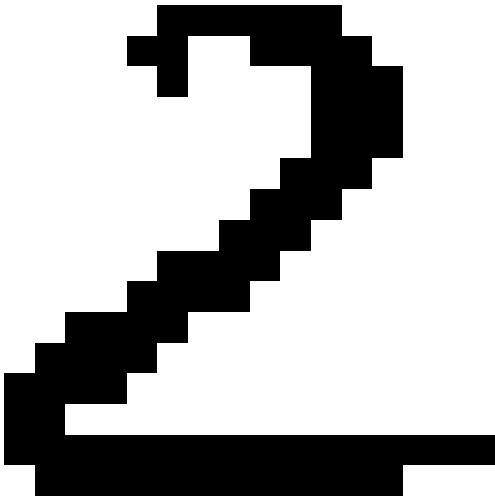
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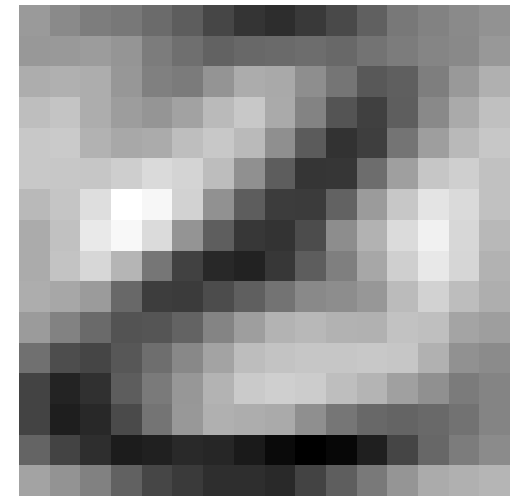
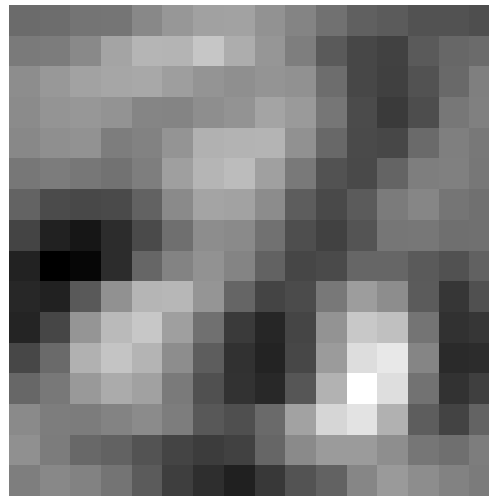
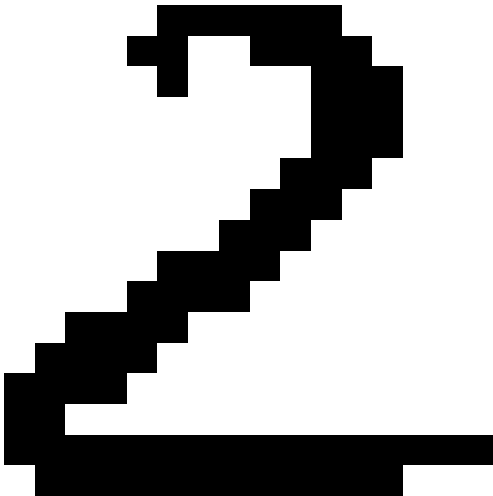
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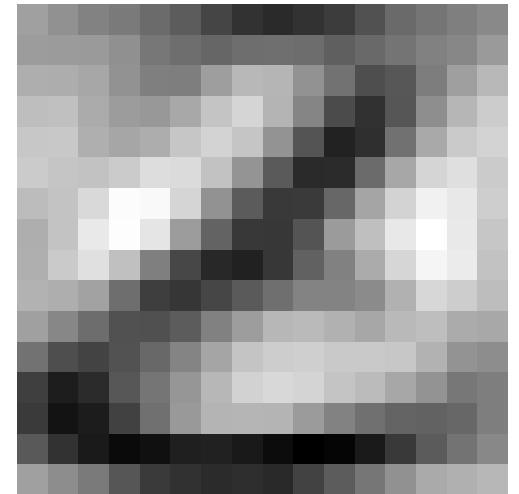
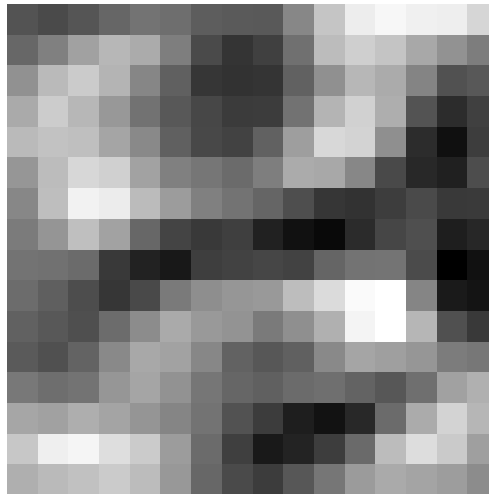
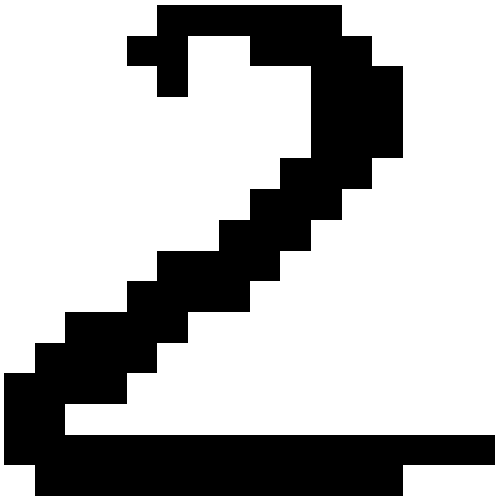
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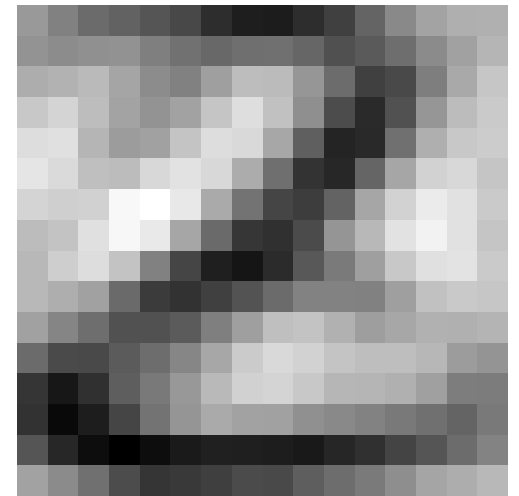
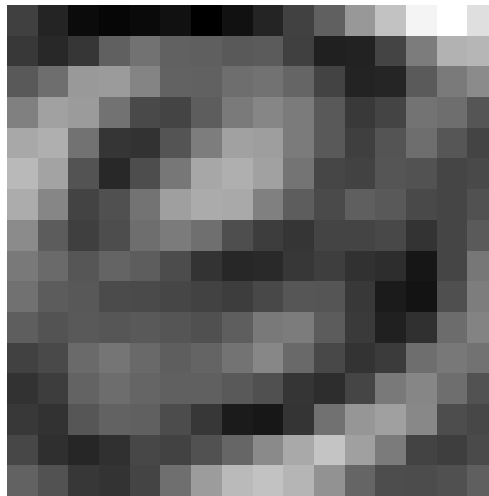
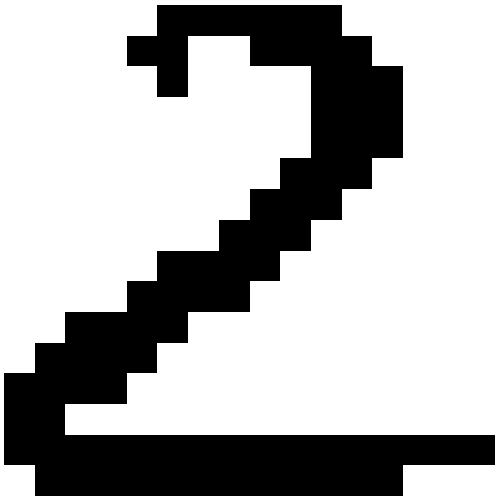
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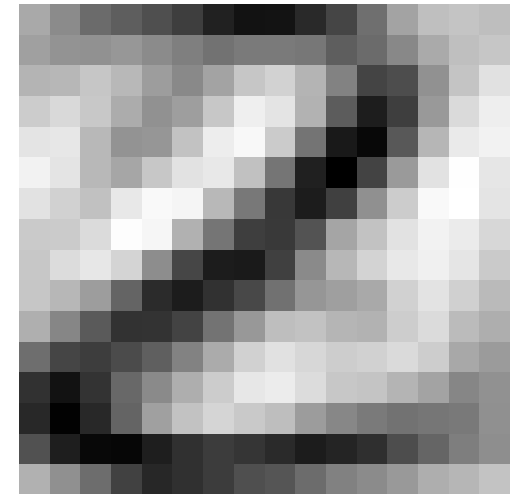
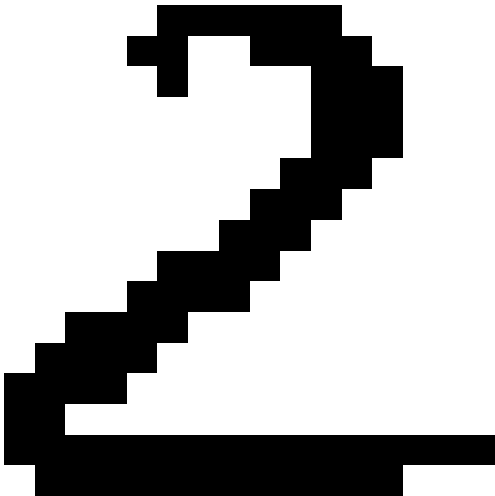
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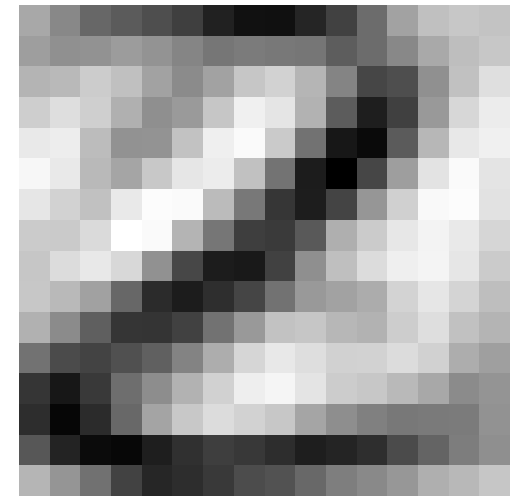
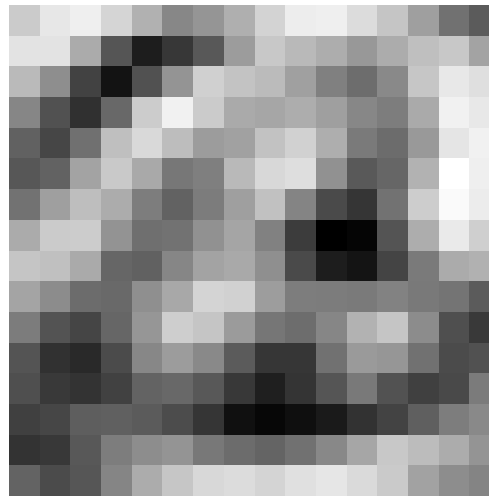
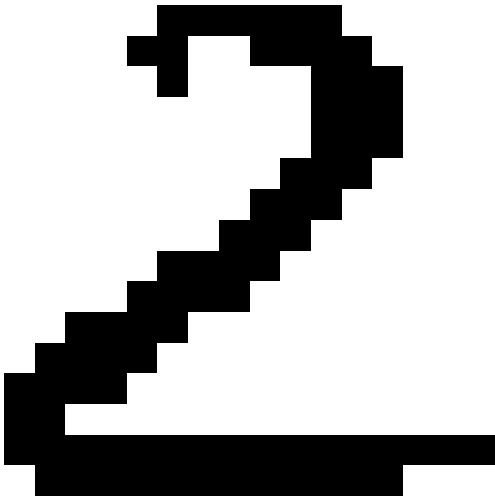
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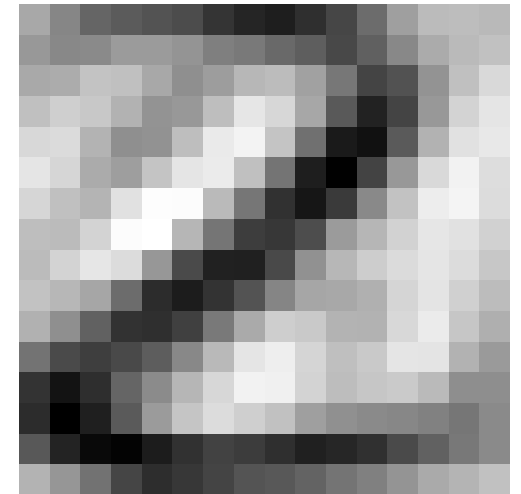
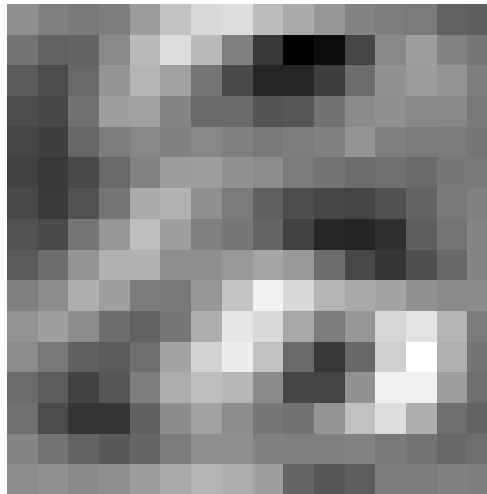
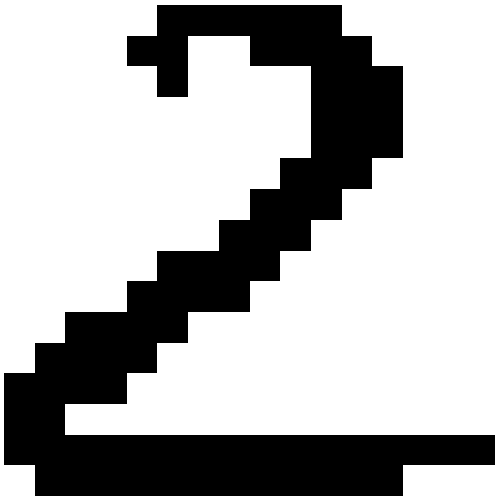
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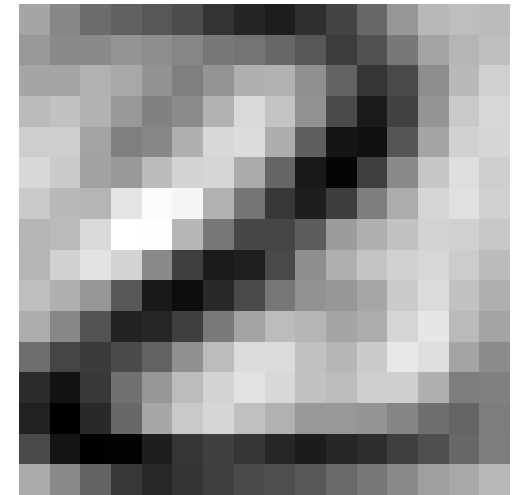
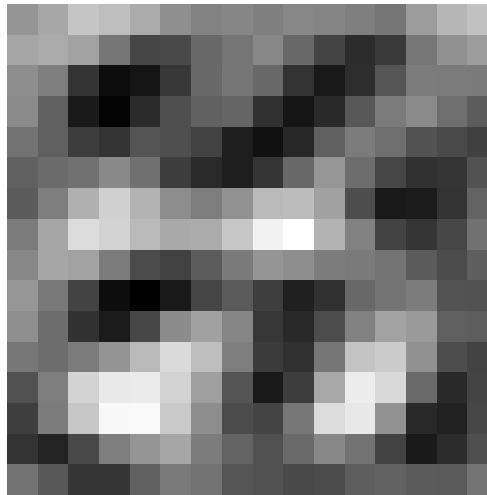
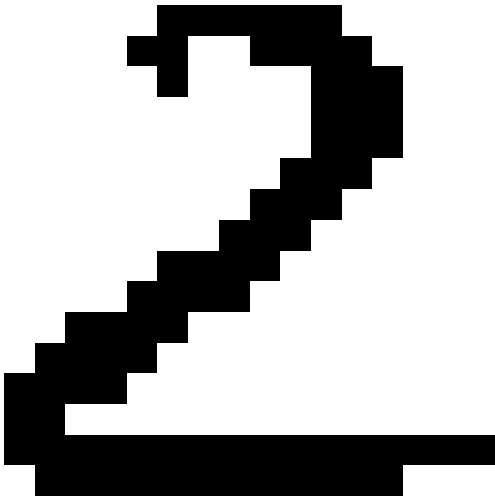
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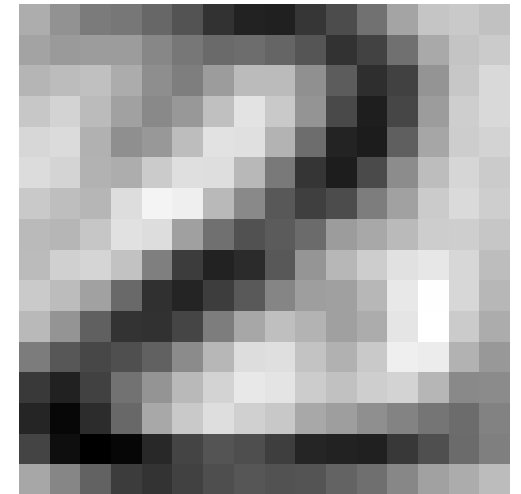
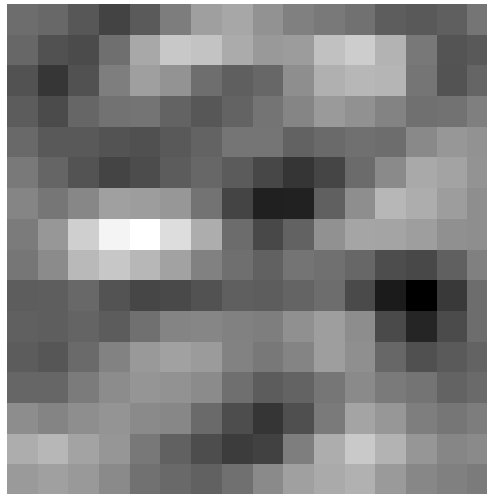
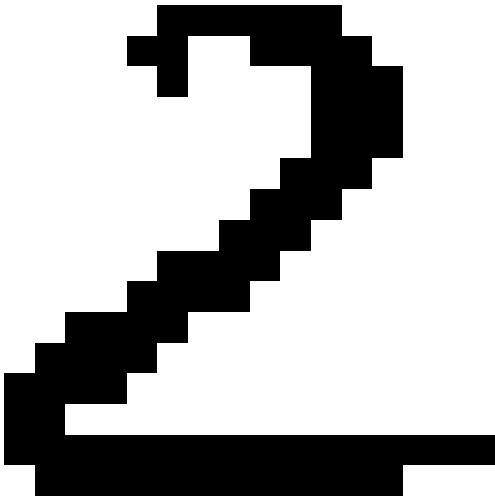
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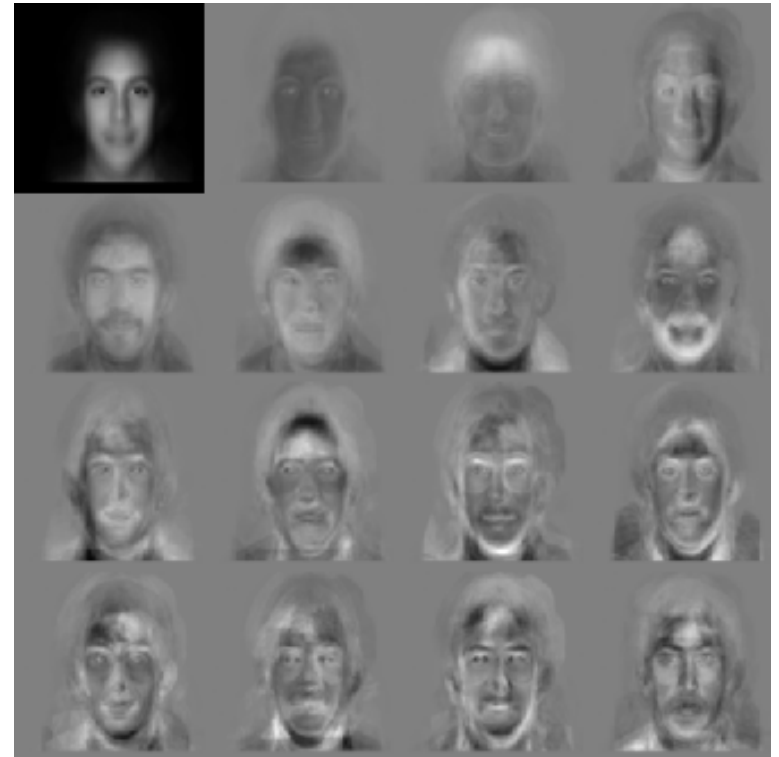
1.6 -1.1 -1.6 2.1 -0.52 2.8 0.72 0.7 -0.68 -0.41 -1.4 -1.5 -0.54 -0.62 1.3 -1.4 -0.27 0.74 0.77 -1





# Outline

1. Covariance Matrices
2. Principal Component Analysis
3. **Duality**



# PCA for Images

- An image often contains around  $p = 256 \times 256 = 64k$  pixels
- In standard PCA we would create an  $p \times p$  matrix with over  $4 \times 10^9$  elements
- This is intractable
- $m$  images span at most a  $m - 1$  dimensional subspace
- Usually this subspace will be much smaller than the space of all images  $m \ll p$

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# Dual Matrix

- The covariance  $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$  is a  $p \times p$  matrix
- Consider the  $m \times m$  matrix  $\mathbf{D} = \mathbf{X}^\top \mathbf{X}$
- Suppose  $\mathbf{v}$  is an eigenvector of  $\mathbf{D}$

$$\mathbf{D}\mathbf{v} = \lambda\mathbf{v}$$

$$\mathbf{X}^\top \mathbf{X}\mathbf{v} = \lambda\mathbf{v}$$

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$$\mathbf{C}\mathbf{X}\mathbf{v} = \lambda\mathbf{X}\mathbf{v} \quad \Rightarrow \quad \mathbf{C}\mathbf{u} = \lambda\mathbf{u}$$

- $\mathbf{u} = \mathbf{X}\mathbf{v}$

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- Matrices  $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$  and  $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$  have the same eigenvalues
- Can use the dual  $m \times m$  matrix  $\mathbf{D}$  to find eigenvalues and eigenvectors of  $\mathbf{C}$
- Note that  $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$  has components  $D_{kl} \propto (\mathbf{x}_k - \boldsymbol{\mu})^\top(\mathbf{x}_l - \boldsymbol{\mu})$
- Takes  $O(p \times m \times m)$  time to construct  $\mathbf{D}$
- We work in a “dual space” which is the space spanned by the examples

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# What Does a Subspace Look Like?

- Consider  $\mathbf{y}^1 = \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}$ ,  $\mathbf{y}^2 = \begin{pmatrix} 8 \\ 6 \\ 2 \end{pmatrix}$  with mean  $\boldsymbol{\mu} = \begin{pmatrix} 5 \\ 5 \\ 3 \end{pmatrix}$
- Subtracting the mean  $\mathbf{x}^i = \mathbf{y}^i - \boldsymbol{\mu}$  we can construct matrix

$$\mathbf{X} = \begin{pmatrix} x_1^1 & x_1^2 \\ x_2^1 & x_2^2 \\ x_3^1 & x_3^2 \end{pmatrix} = \begin{pmatrix} -3 & 3 \\ -1 & 1 \\ 2 & -2 \end{pmatrix}$$

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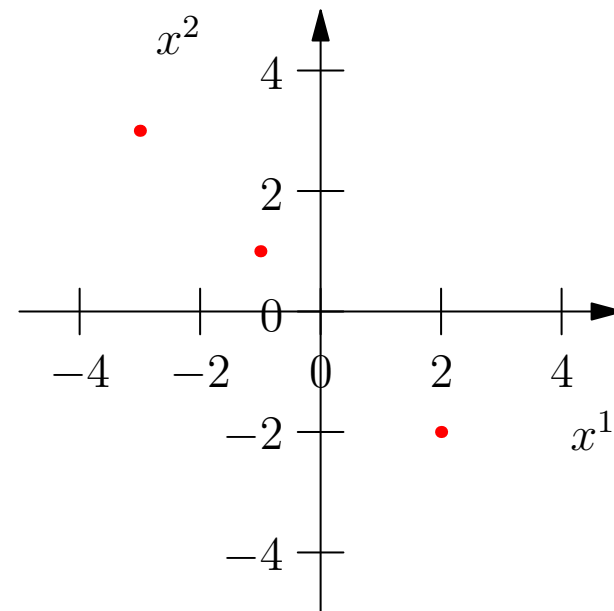
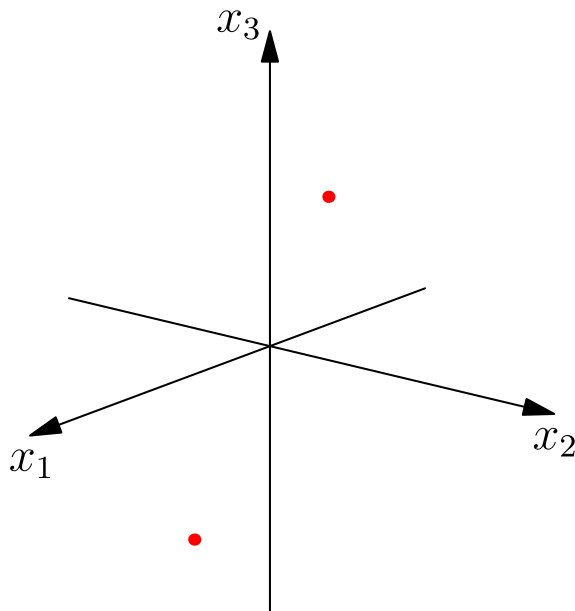
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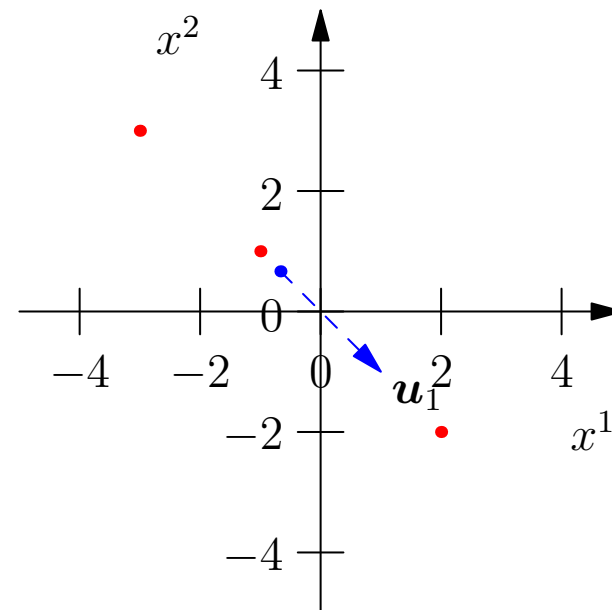
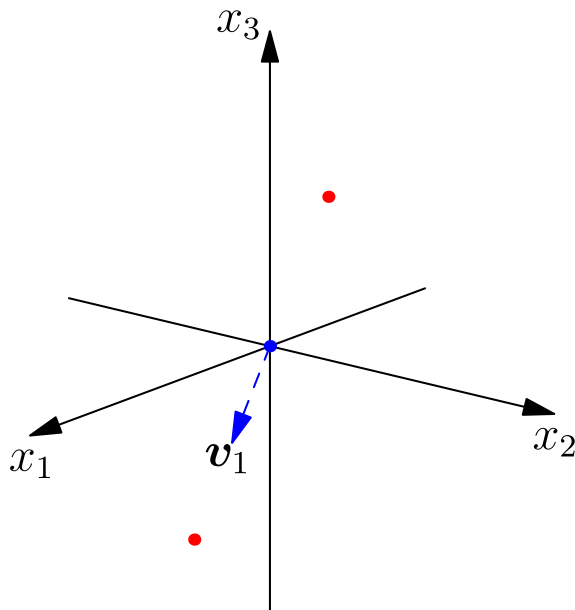
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# Summary

- PCA allows us to reduce the dimensionality of the inputs
- We project the inputs into a sub-space where the data varies the most
- We can work in either the original space ( $\mathbf{X}\mathbf{X}^T$ ) or the dual space ( $\mathbf{X}^T\mathbf{X}$ )
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- We will see examples of dual spaces again when we look at SVMs